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INDUSTRY RESEARCH REPORT · NOVEMBER 2026

Sprayable coatings for agricultural film

Spray-on coatings turn commodity film into functional covers: global trends, products and technical specifications, raw materials, production, application methods and investment data.

Shading · NIR-reflective · diffusing · anti-drip · emerging functions

Edition 1.0

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About this report. Compiled in November 2026 from public sources: manufacturer technical data sheets, patents, peer-reviewed studies, official statistics, trade press and market-research summaries. Inline links point to each source. Market-research figures are publisher teasers with undisclosed scope; derived estimates (market size, plant economics, cost per area) are labelled as such and rest on stated assumptions. Product performance values are mostly manufacturer laboratory data. Nothing here is investment advice; verify figures and run freedom-to-operate and regulatory checks before committing capital.

CHAPTER 1

Executive summary

Today, three kinds of spray products can turn a normal PE, EVA or PO (common plastic) film for greenhouses, tunnels or mulch into a cover with extra functions. The first kind goes on the outside and controls light and heat: white shading, NIR-reflective coatings (NIR is near-infrared, the invisible heat part of sunlight) and light-diffusing coatings that spread light. The second kind goes on the inside and stops water drops from forming (anti-drip or anti-fog). The third kind is the removers that wash the outside coatings off, so growers can use them for one season. Other wished-for functions are either built into the film in the factory or still in the laboratory: UV control added after production, anti-dust or antistatic, light conversion, and shading that changes with temperature (thermochromic). The main products use simple, cheap, well-known chemistry. Removable shading concentrates contain 30–95 wt% CaCO3/TiO2 (chalk and white pigment) in an acrylic or starch binder that dissolves in alkali. Anti-drip liquids contain 2–12 wt% colloidal silica (very small silica particles in water) in a binder that likes water. Factories only blend and fill, and they cost about €350–550 of capex (building investment) per tonne of yearly capacity. The market is a small, concentrated niche inside a large film industry. Agricultural films are worth about **US$12.5–14.3 bn (2025)** and grow 6–9 %/yr, but no market publisher measures spray-on coatings. Our own bottom-up estimate puts the materials market at **€60–370 m, with a central estimate of about €170 m**. Lumiforte leads it; it is a merger of Mardenkro/ReduSystems and Sudlac, owned with private equity (PE). Hermadix/Royal Brinkman is next, and very cheap simple chalk paints are common in China. For growers the economics are good. Basic shading costs about €0.02–0.18/m² per application, and independent tests show 2–5 °C cooling and 2–6 % more yield from diffuse light. For a producer, the barrier is not money for a plant. It is access to sales channels, proof of performance in the field, and fit with rules: bans on discharging rinse water, the EU rules on microplastics and PFAS, and film-recycling standards. There are three clear open opportunities: an anti-drip renewal spray that works on all film types for Asia's ageing PE/PO films, shading products made for drones with standard methods, and a mid-priced NIR-selective product between Chinese chalk and expensive European products.

Agricultural film market, 2025

US$12.5–14.3 bn

Fortune BI, Mordor, GMI; 6–9 %/yr

Spray-on coating materials (derived)

≈€170 m

range €60–370 m; bottom-up estimate

Mapped greenhouse area, 2019

≥1.3 Mha

60 % in China; Tong et al. 2024

Independently measured cooling

2–5 °C

$CaCO_3$ whitewash, Almería trials

Key takeaways

- **Sprayable today:** exterior shading, NIR-reflective and diffusing coatings; interior anti-drip coatings; and the removers that make exterior coatings seasonal.
- **Still built into the film:** UV control, anti-dust/antistatic and light conversion (sprayed versions are lab or pre-commercial).
- **Chemistry is simple:** $CaCO_3$/TiO_2 in alkali-soluble acrylic or starch binders; 2–12 wt% colloidal silica for anti-drip. Core patents have expired.
- **Plants are cheap** (≈€350–550 capex per tonne of annual capacity); market access, independent field data and regulatory fit are the real barriers.
- **White spaces:** cross-film anti-drip renewal spray for Asia, drone-native shading concentrates with standard protocols, and a mid-priced NIR-selective grade.

CHAPTER 2

Three functions can be sprayed today; four are still mostly built into the film

The table below shows the seven functions in the brief and what growers can really buy as a spray product in 2026. The split is very clear.

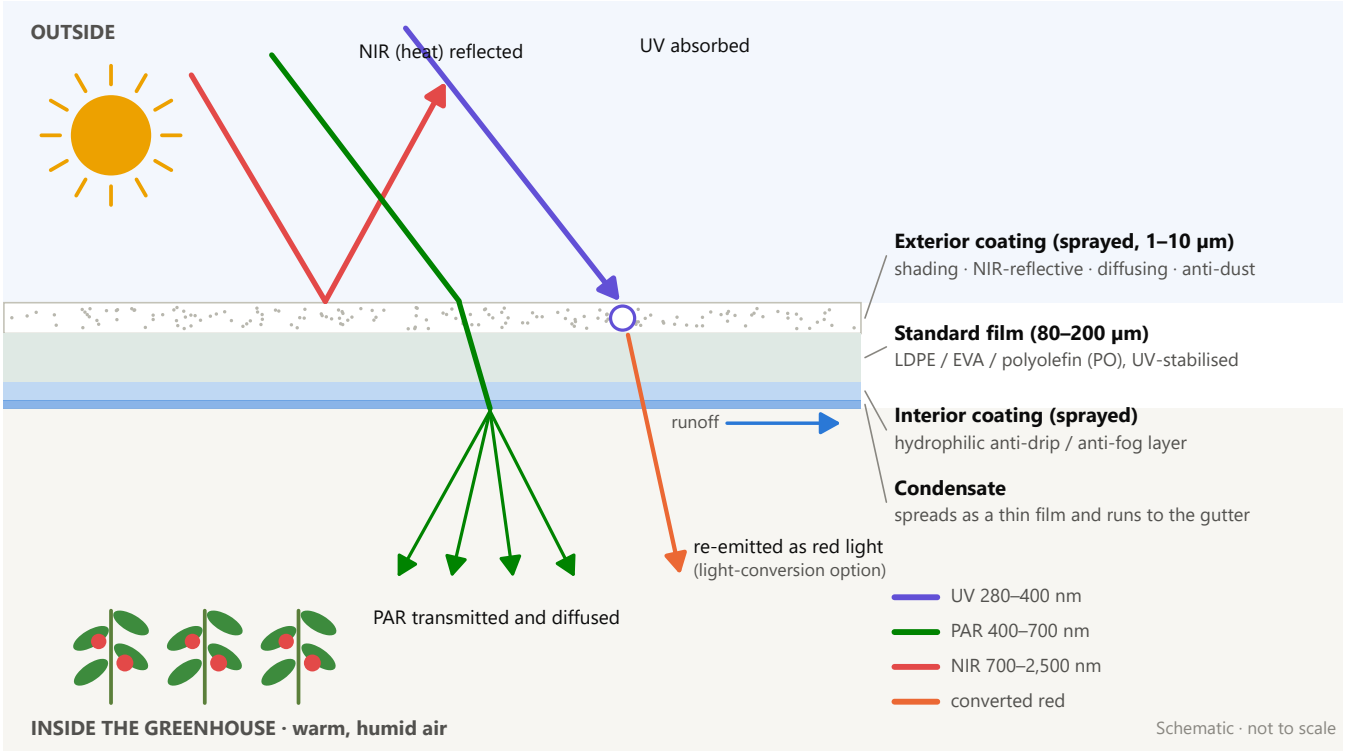


Figure 1. How a sprayed functional film works. Exterior coatings act on incoming sunlight; an interior hydrophilic coating controls condensation.
Schematic by the report authors; layer thicknesses not to scale.

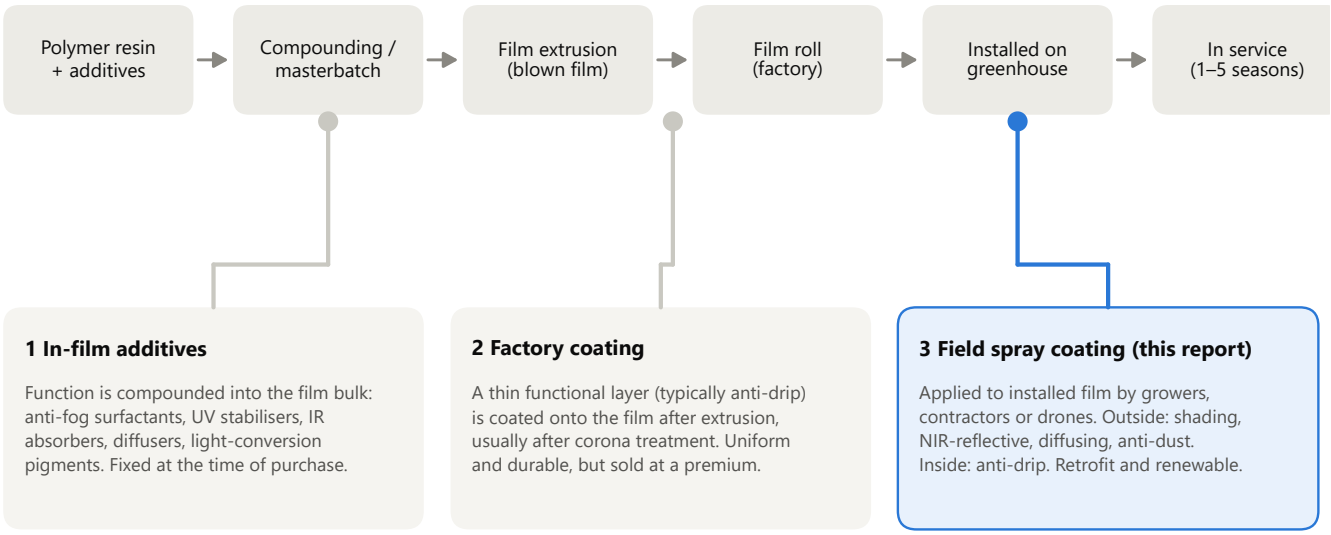


Figure 2. Three routes to a functional film. Field spray coating is the only route that can retrofit, renew or remove a function after installation.
Report authors.

- **Shading/cooling, NIR-reflective and light-diffusing coatings** go on the outside of the cover. They are a fully commercial group (TRL 9; TRL means Technology Readiness Level, from 1 = idea to 9 = on sale). There are dozens of products.
- **Anti-condensation coatings** go on the inside. They are on sale, but there is little independent proof.
- The other functions exist mainly as additives inside the film, or as research prototypes.

The products researcher found no spray product for UV blocking, photocatalytic self-cleaning (cleaning by a light-driven reaction) or antistatic use on greenhouse film. Light-conversion sprays are not yet on sale (ReduSystems downloads; Univ. of Bath).

Table 1. Functions, how they work, and market status of spray-on products (2026)

Function	How it works	Typical active materials	Spray product status	Example products	Maturity (TRL, est.)
White shading / cooling	Particles with high refractive index and air gaps scatter PAR and NIR back	CaCO3, TiO2, kaolin, lime	On sale; removable or wears off by itself	ReduSol, Eclipse LD/F2/F4/F6, Hermadix Q3/Q4, Kool Ray, Chinese 降温剂, Korean 차광도포제	9
NIR-reflective ("heat-blocking")	Reflects 780–3,000 nm; lets PAR pass	Interference pigments (TiO2–SiO2 on mica), other secret pigments	On sale, high price	ReduHeat, Transpar, Q-Heat, D-Gree	9
Light-diffusing	Particles about the size of the light wavelength scatter light forward (Mie scattering); haze rises with layer thickness	Scatterers that absorb little light	On sale	ReduFuse, ReduFuse IR, Optifuse/IR, D-Fuse/Extra	9
Anti-drip / anti-fog (inside)	Raises surface energy, so water spreads as a clear layer, not drops	Colloidal silica/alumina plus binder; surfactants	On sale but niche; lasts one season to a few years	AntiCondens Plastic/Glass, Sun Clear, Nurutobodon (JP)	9 (spray); 4–5 (next-gen, chemically bonded)
Anti-reflection	Porous layer with low refractive index	Not disclosed	On sale, **glass only**	AntiReflect	9 (glass)
Anti-dust / self-cleaning / antistatic	Light-driven surface that loves water, or a surface that strongly repels water	Anatase TiO2/SiO2; siloxanes	Roof coating with light-driven cleaning launching in NL with no data; no film product found	Multipurus (NL)	3–4 on film; 7–8 claimed on roofs
UV control	Absorbs 280–400 nm to confuse insect vision or protect the film	Benzophenone/benzotriazole absorbers, HALS	Sold as **film additive**; no spray product found	UV-cut films (e.g., Nippon Matai)	9 (additive)
Light conversion	Down-conversion: UV/blue/green light is sent out again as red	Eu3+ complexes, Mn4+ phosphors, quantum dots, carbon dots	Spray versions in lab or not yet on sale; films are on sale	Lambda Agri spray (pre-commercial), UbiGro QD film	4–5 (spray); 8 (QD film)

Sources for Table 1: product data sheets (TDS) and labels linked in later sections; AIP Advances 2025 review; HortiDaily 2026 Multipurus; Frontiers in Plant Science 2025 review.

2.1 Each coating works by changing how light or water behaves at the surface

All these coatings are like "switches" for light or water, added to a cheap polyolefin (PE-type) film. Anti-drip is needed because LLDPE has a low surface energy of only about 30 dyn/cm. So water forms round drops on it, and these drops scatter light. Drops with a contact angle (how much the drop stands up from the surface) below **48.8°** do not trap light by total internal reflection. When the angle is close to zero, light transmission is almost unchanged (AIP Advances review).

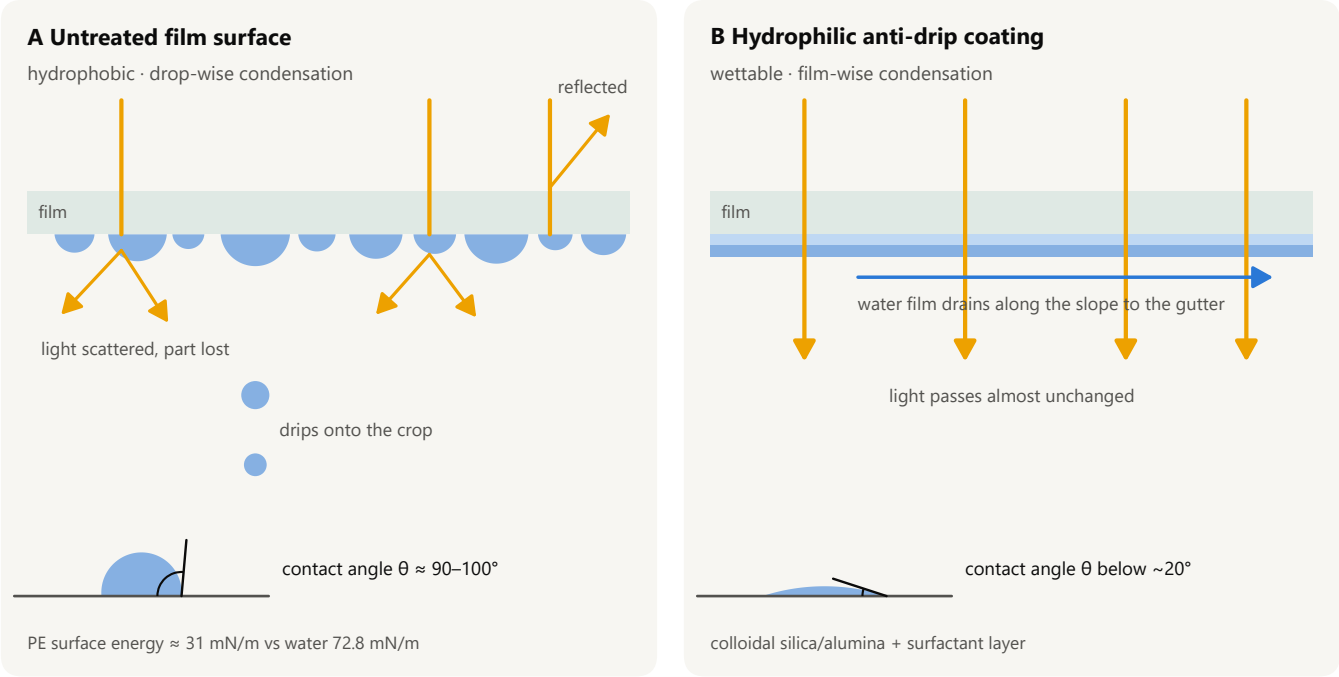


Figure 3. Drop-wise versus film-wise condensation. Droplets with high contact angles scatter and reflect light and drip onto the crop; a hydrophilic coating spreads condensate into a thin draining film.

Schematic by the report authors; values from the AIP Advances 2025 review and standard surface-energy data.

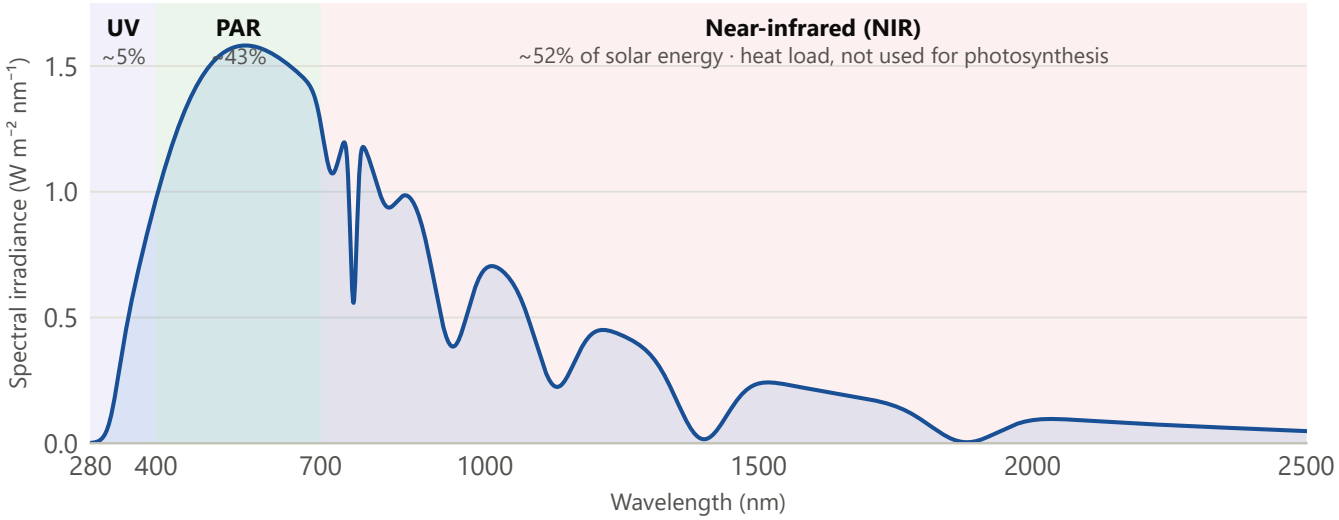


Figure 4. What each coating targets in the solar spectrum. Roughly half of solar energy is near-infrared heat that plants cannot use; NIR-selective coatings aim to reject it while passing PAR.

Approximate clear-sky spectrum computed by the report authors (schematic, not measured data).

A lot of light is at stake:

- A patent says drops on greenhouse film lose **15–30** % of light (WO2022023782A1).
- Wageningen calculations, quoted by the market leader, say condensation loses up to 9 % on glass. The seller says up to 15–20 % on plastic (HortiDaily 2019).

Shading, diffusing and NIR coatings work by scattering light. Rutile TiO2 has a refractive index of 2.6, and crystals of ~220 nm reflect the most visible light (Wikipedia: TiO2). Interference pigments use thin layers with high and low refractive index. This puts their reflection in the NIR, while they look almost colourless (US 7410685 B2). The industry defines the heat band as **780–3,000 nm** and PAR (photosynthetically active radiation, the light plants use to grow) as **400–700 nm**. Water in the crop absorbs IR strongly above 1,200 nm, but plants do not use it for photosynthesis (HortiDaily).

CaCO3 whitewash has one useful feature: it becomes more transparent when wet. So light levels partly follow the weather (EP0999736A1). The chemistry researcher thinks water fills air gaps in a pigment layer that has little binder on purpose. This is a guess, not a measured fact. A second design point concerns NIR absorbers such as CsxWO3 and ATO. They turn NIR into heat inside the cover. So for cooling, pigments that reflect are physically better than pigments that absorb.

CHAPTER 3

Data sheets show that shading, NIR and diffusion can be set by the dose

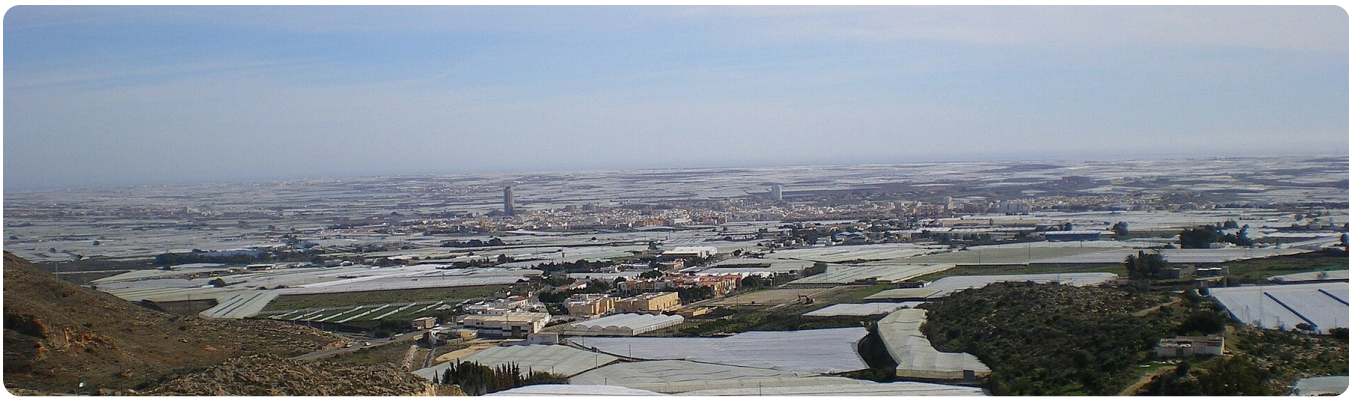


Photo: ANE · CC BY 3.0 · Wikimedia Commons

The outside coatings work like a dial. The grower chooses a dose (buckets per hectare, or a dilution ratio). He then gets a predictable level of PAR shading, NIR shading and scatter. Lumiforte publishes the most complete data. Its TDS (technical data sheets) give hemispherical (all directions) and perpendicular (straight-through) net shading for PAR and NIR. They also give Hortiscatter (a measure of light spreading) for the diffusing products. Hermadix publishes dose against shading measured in lux or PAR, and lifetime in weeks.



Figure 5. Aerial view of a dense patchwork of plastic-film greenhouse roofs near El Ejido, Almería, Spain; many roofs appear whitened.

Photo: kallerna · CC BY-SA 4.0 · Wikimedia Commons



Figure 6. Astronaut photograph from the International Space Station of the plastic greenhouses of the Campo de Dalías, south-west of Almería, Spain (2004).

Photo: NASA astronaut (ISS Expedition crew), via NASA Earth Observatory · Public domain · Wikimedia Commons

Be careful when comparing figures, because they use different bases:

- Since 2020 Mardenkro reports **net** shading for the coating alone.
- **Gross** shading adds about 17 % for glass and 10–20 % for structure. So a 60 % gross figure means about 30 % from ReduSol alone (HortiDaily).
- North American catalogue values are about half of the TDS laboratory maximum. ReduSol at 4–5 buckets/acre is quoted at 22–25 % net (Griffin catalogue). The TDS gives 37–40 % at 10 buckets/ha.
- Hermadix Q3 reports shading as "% in lux", which is not based on PAR (Q3 sheet).

You can compare values from different makers only after you put them on the same basis.

3.1 Main comparison of the leading spray products

Table 2. Technical comparison of major spray-on products (maker laboratory values unless stated)

Product (maker)	Function	Look, pH, density, viscosity	Pack	Machine dose	Coating shading	NIR / scatter	Lifetime	Removal	Surfaces
ReduSol (Lumiforte/ReduSystems)	White shading	White, pH 9, 1.4 g/cm³, 7,800 mPa·s	20 kg (14.2 L)	5–25 buckets/ha	Hemi net PAR 31–55 %; gross 43–63 %	NIR 25–56 %	4–8 months	ReduClean ≥250 L/ha	Glass, acrylic, PC, film
ReduHeat (ReduSystems)	NIR-reflective	Beige, pH 9, 1.1 g/cm³, 7,600 mPa·s	15 kg (13.5 L)	15–25	Hemi net PAR 22–42 %	NIR 32–55 %	Season	ReduClean	Glass, acrylic, PC, film
ReduFuse (ReduSystems)	Diffuse	White, pH 8.5, 1.0 g/cm³, 7,300 mPa·s	15 kg (14.9 L)	12–14	Hemi net 9–10 %	NIR 4 %; Hortiscatter 41–49 %	Season	ReduClean diffuse	Glass, plastic (label varies)
ReduFuse IR (ReduSystems)	Diffuse + NIR	Beige, pH 9.6, 1.1 g/cm³, 8,000 mPa·s	15 kg (13.5 L)	10–22	Hemi net 14–30 %	NIR 15–32 %; HS 53–86 %	Season	ReduClean diffuse	Glass, acrylic, PC, film
Transpar (Sudlac)	NIR + diffuse	Beige, pH 6.5–8, 1.1 g/cm³, 2,200 mPa·s	15 kg (14.2 L)	15–25	Perp net PAR 10–24 %; gross 32–44 %	NIR 22–44 %	Season	Topclear 240 L/ha	All incl. film
Optifuse IR (Sudlac)	Diffuse + NIR	Milky, 1.04 g/cm³, 2,400 mPa·s	15 kg (14.4 L)	10–15	Perp net 7–13 %	NIR 16–22 %; HS 51–75 %	Season	Topclear Fuse	Glass, plastic
Eclipse LD / LD+ (Sudlac)	White, removable	White, pH 8–9, 1.35–1.45 g/cm³, 6,500 mPa·s	20 kg	10–30	Perp net 29–65 % (LD+ 32–68 %)	NIR ≈ PAR	4–8 months	Topclear	All; LD+ ready for drone/helicopter
Eclipse F2/F4/F6 (Sudlac)	Wears off by itself	Beige, pH 8.8–9.8, 1.43 g/cm³, 4,300 mPa·s	20 kg	5–30	Perp net 17–65 %; gross 40–76 %	NIR ≈ PAR	≤2 / 2–4 / 4–6 months	Wears off	All
Canope F4 (Sudlac)	Biodegradable, temporary	Beige, modified corn-starch binder, OECD 301F	20 kg or 1,400 kg IBC	7–16	Net 32–50 %; gross 49–67 %	–	≤4 months	Wears off	Acrylic, PC, film (not glass)
Q3 / Q4 White (Hermadix)	White, CaCO3	–	20 kg	10–34	40–80 % (lux)	–	4–25 weeks	Removit	Glass, film, PC, acrylic
Q-Heat (Hermadix)	NIR + diffuse	–	15 kg	17–44	PAR 15–40 %	NIR 40–70 %; haze 90–95 %	10–24 weeks	Removit	Same
D-Fuse / D-Fuse Extra / D-Gree (Hermadix)	Diffuse (± NIR)	–	15 L	13 machine / 17 manual	2–3 % / 9–11 % / 15–20 %	Haze 86 / 95 / 93 %; D-Gree NIR 46 %	16–20 weeks	Removit or water rinse	Glass, film, PC, acrylic
Kool Ray Classic (Continental)	White, wears off	–	1, 5, 55 gal	1:8 dilution	~30 % less light	–	Season	Weathering	Mainly glass, with binder

Product (maker)	Function	Look, pH, density, viscosity	Pack	Machine dose	Coating shading	NIR / scatter	Lifetime	Removal	Surfaces
AntiCondens Glass/Plastic (ReduSystems)	Inside anti-condensation	Opalescent, 1.1 g/cm³, 3 mPa·s	20 L	60 L in ~1,000 L/ha	Light +5–7 % glass, +15–20 % plastic (claim)	–	1 season to ≥1 year	Wears off; cleaners remove it	Glass or film
Sun Clear (maker not confirmed)	Inside anti-condensation	Water-based surfactant	473 mL – 1 gal	100:1 plastic, 50:1 glass	"Up to 50 %" more light (claim)	Antistatic claim	2–5 years (claim)	10:1 vinegar	Film, rigid, glass
AntiReflect (ReduSystems)	Anti-reflection	pH 10.2–10.7, 3.86 mPa·s	Ready to use	1,200 L/ha	+2.5–3 % transmission	–	2 years	GS-4	**Glass only**

Sources: ReduSol TDS; ReduHeat TDS; ReduFuse TDS; ReduFuse IR TDS; Transpar TDS; Optifuse IR TDS; Eclipse LD TDS; Eclipse F4 TDS; Canope F4 TDS; Royal Brinkman Q-series; Royal Brinkman diffuse coatings; Griffin Kool Ray; AntiCondens product sheet; Greenhouse Megastore Sun Clear; AntiReflect TDS.

3.2 Dose-response curves give data ready for charts

The dose-response data show the key business point. **NIR products block more heat for each unit of PAR lost.**

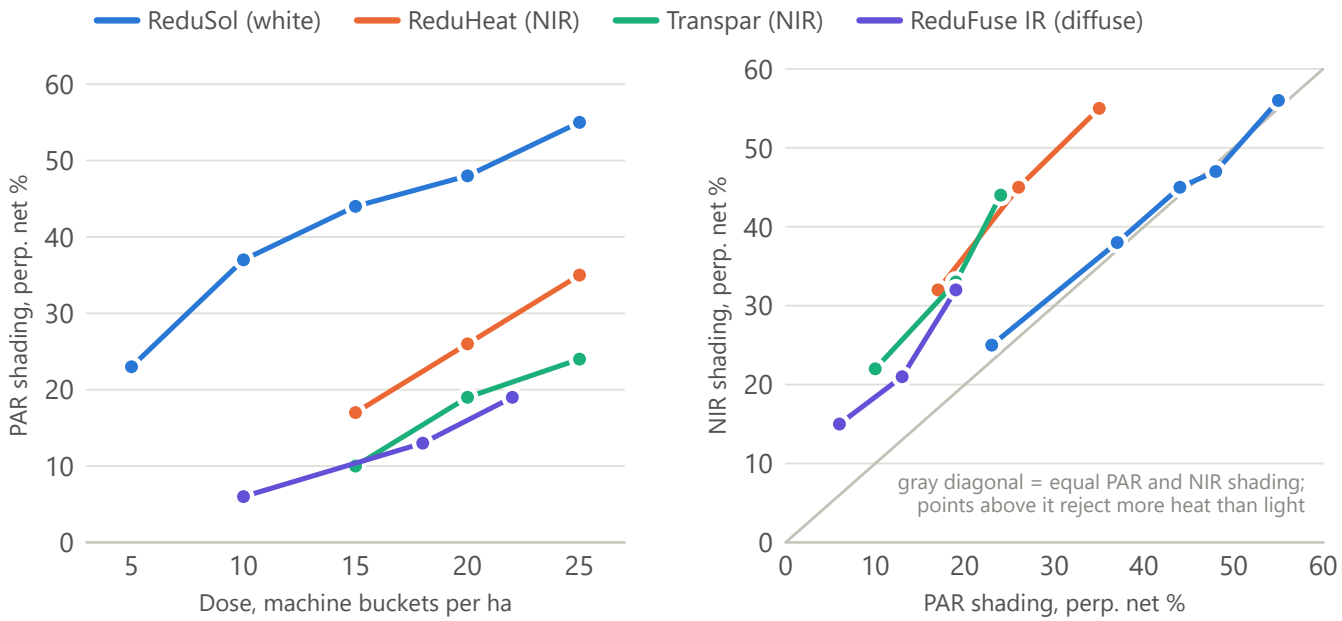


Figure 7. Dose response of four commercial exterior coatings. Left: PAR shading rises with dose. Right: NIR grades sit above the diagonal, rejecting more heat per unit of light lost than white shading.

Source: Lumiforte technical data sheets (perpendicular net values, manufacturer laboratory data); Table 3.

- At 25 buckets/ha, ReduSol removes 55 % of PAR and 56 % of NIR.
- ReduHeat at the same dose removes only 35 % of PAR (perpendicular) but 55 % of NIR.
- Diffusing products lose only a few percent of PAR, but scatter 40–86 % of it.

Haze depends mainly on how thick the coating is (HortiDaily).

Table 3. Dose-response (machine buckets/ha, manual in brackets; values in %)

Product	Dose	Hemi net PAR	Gross PAR	Perp net PAR	Perp net NIR	Hortiscatter
ReduSol	5 (6)	31	43	23	25	–
ReduSol	10 (11)	40	50	37	38	–
ReduSol	15 (17)	45	54	44	45	–
ReduSol	20 (22)	48	57	48	47	–

Product	Dose	Hemi net PAR	Gross PAR	Perp net PAR	Perp net NIR	Hortiscatter
ReduSol	25 (28)	55	63	55	56	–
ReduHeat	15 (17)	22	36	17	32	–
ReduHeat	20 (22)	32	44	26	45	–
ReduHeat	25 (28)	42	52	35	55	–
ReduFuse	12 (13)	9	25	2	4	41
ReduFuse	14 (15)	10	26	3	4	49
ReduFuse IR	10 (11)	14	30	6	15	53
ReduFuse IR	18 (20)	24	37	13	21	77
ReduFuse IR	22 (24)	30	42	19	32	86
Transpar	15 / 20 / 25	–	32 / 40 / 44	10 / 19 / 24	22 / 33 / 44	–
Optifuse IR	10 / 15	–	–	7 / 13	16 / 22	51 / 75
Eclipse LD	10 / 20 / 30	–	47 / 57 / 72	29 / 44 / 65	29 / 44 / 65	–
Eclipse F4	5 / 10 / 20 / 30	–	40 / 51 / 67 / 76	17 / 34 / 56 / 65	= PAR	–
Canope F4	7 / 11 / 16	32 / 38 / 50 (net)	49 / 55 / 67	–	–	–

Sources: the Lumiforte TDS linked in Table 2. The Eclipse LD+ TDS gives 67 % NIR at 10 buckets/ha, which is higher than its 20-bucket value. This is probably a typing error (Eclipse LD+ TDS).

Public data from Asia show the same logic with cheaper chemistry. Korea's RDA (Rural Development Administration) lime-and-starch coating gives about 50 % shading at 1:4, 30 % at 1:7 and 15 % at 1:10 (농업인신문). At 4–10× dilution it reduces total light by 3–11 %, PAR by 4–12 % and far-infrared by **21–63** %. It lowers daytime air temperature by 1.0–4.3 °C (농사로). In Almería, PAR transmission falls exponentially as the CaCO3 dose rises. It falls in proportion below about 50 g/m², and only a little more between 50 and 100 g/m² (Moreno-Teruel et al. 2020). Above 0.5 kg/L, cover transmission falls by more than 50 % (López-Martínez et al. 2019).

3.3 Anti-drip, anti-reflection and the missing functions

Inside anti-condensation sprays are the only proven "film renewal" products for plastic.

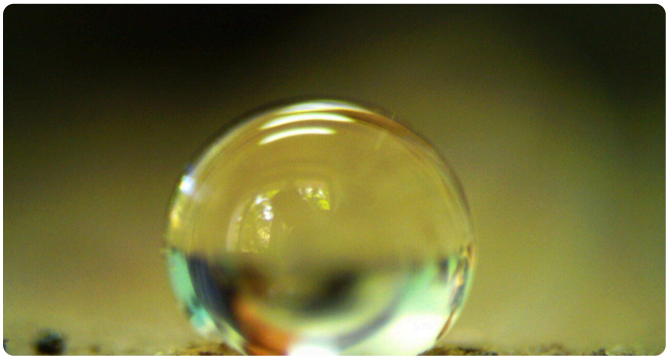


Figure 8. Water droplet beading on a highly hydrophobic surface (pelletised hydrophobic sand), contact angle about 145 degrees.

Photo: Murat Baisariyev and Aidar Seralin · CC BY-SA 4.0 · Wikimedia Commons



Figure 9. Interior of a multi-span plastic-film tomato greenhouse in South Korea, with young plants on black mulch beds.

Photo: Choi Kwang-mo · CC0 · Wikimedia Commons

- **AntiCondens** forms a layer that likes water. There are separate versions for glass and plastic. Growers use about 60 L of product in about 1,000 L/ha of spray liquid. Its label carries H319 (eye irritation warning) and it contains CMIT/MIT preservatives (AntiCondens Plastic label).
- **Sun Clear** is a surfactant liquid sprayed with a mister, with the nozzle within 100 mm of the film. Its seller claims 2–5 years per application (First Tunnels).
- In Japan, Mikado Kakou sells the two-part kit **Nurutobodon** (4 L A + 2 L B in 200 L water for about 400 m²). It renews anti-drip performance, but only on the maker's own PO films (Mikado Kakou).
- **AntiReflect** adds 2.5–3 % transmission for two years, but only on glass (AntiReflect TDS).

For the other functions, there is little evidence. The only anti-dust or antistatic claim for a spray is a seller's statement that Sun Clear "reduces static charge" (First Tunnels). UV control is only a side benefit. ReduSol is said to block part of the UV and make plastic film last longer (HortiDaily 2017). GRAB says whitening protects plastic covers from damage by light (GRAB 2020). For light conversion, the closest product is the sprayed varnish from Lambda Agri/Bath/Cambridge. It turns UV into red light and gave **+9 % basil yield**. A patent is pending, and sales are expected "in a few years" (Univ. of Bath).

3.4 Test standards are written for films, not for sprays

Coating optics are mostly measured at the WUR Lightlab (Wageningen University). Haze means light that is bent by more than 2.5°. NEN 2675 (2018) replaces haze with Hortiscatter and covers PAR, UV and NIR transmission (HortiDaily).

Anti-drip standards are written for films:

- The GB 4455 test uses a water bath at 60 ± 1 °C, with the film tilted at 15°. The film fails when drops cover ≥30 % of the area, or dripping covers ≥50 % (Sohu repost of standard).
- China's 2023 draft group standard for coated PO film adds 60 °C and 30 °C baths, because coated films behave differently. It sets a starting contact angle of **≤35°** (CPPIA draft; ttbz.org.cn).
- The new **EN 13206:2025** for PE/EVA films defines non-thermal, thermal clear and thermal diffusing types. It does not mention anti-drip (Genorma).
- Korea's KS standard also has no anti-drip rule (농업인신문).

We found no country with a standard for anti-drip or shading products applied in the field. This is an open space with no entry barrier. But it also means there is no accepted performance benchmark.

CHAPTER 4

Recipes use cheap minerals, alkali-soluble acrylics and colloidal silica

Makers do not disclose their recipes. No TDS gives the solids, binder or pigment chemistry (ReduSol TDS). But European and Japanese patents, read in full, show the basic designs. They come down to three basic recipes.



Figure 10. Titanium dioxide powder dispersing in a glass of water.
Photo: Pixelmaniac pictures · CC0 · Wikimedia Commons



Figure 11. Pile of white calcium carbonate filler masterbatch pellets for plastics production.
Photo: Gabriele85 · CC0 · Wikimedia Commons



Figure 12. Translucent polyethylene pellets next to a centimetre ruler.
Photo: Lluis tgn · CC BY-SA 3.0 · Wikimedia Commons

4.1 Three basic recipe designs cover the market

Removable shading concentrates come from two patents: EP 0999736 (RSB, the company before Mardenkro, 1997) and EP 2045302 (Fenzi, 2007). They contain **30–95 wt% pigment** (CaCO3, TiO2, silicates, gypsum, barite) in **4–60 wt% binder**. In RSB's version, the binder is a vinyl polymer with acid value 40–250. In Fenzi's version, it is a mix of a water-resistant acrylic and an acrylic that dissolves in alkali. It is neutralised with ammonia, so it dissolves again when an alkaline cleaner is sprayed on (EP 0999736 B1; EP 2045302 B1).

Fenzi's Example 1 shows the recipe:

Component	Parts by weight
CaCO3	634
NeoCryl FL711	143
NeoCryl B817	19
Water	~190
Ammonia	3.8
Xanthan-type thickener	1.3
Biocide, defoamer, wetting agent	small amounts

The concentrate is diluted 1:2–1:5. Rain cannot wash it off after about 30 minutes at 20–25 °C. A dry layer of 2–9 µm blocks **15–40** % of light (EP 2045302 B1). The bio-based version uses modified corn starch instead of acrylic. Sudlac's Canope F4 and Lumiforte's LDC 101 came out in 2024 (HortiDaily). Korea's RDA coating uses lime plus biodegradable starch (농민신문 2017). Almería's "Flex"-type products are about 97 % very fine CaCO3 (average 2.8 µm) with less than 3 % glue (López-Martínez et al. 2019).

Colloidal anti-drip liquids are best described in JP 6833585 B2 (2021). Its recipe:

Component	Amount
Colloidal silica, 40–60 nm preferred (Nissan Snowtex grades named)	2–12 wt% of the liquid
Binder: PVA or acrylic, best a polycarbonate PU dispersion with silanol groups	0.5–5 wt%
Surfactant: polyether-modified silicone or acetylenic diol	1–15 phr (parts per hundred) on solids
Water ± alcohols	the rest

The worked example is 4 % silica, 2 % binder and 94 % water. It is dried at 70 °C for 1 minute to a 1.0 µm layer. The patent also claims a "smart" rough surface. It is hazy when dry (10–50 % haze), which prevents leaf burn in summer. It is clear when wet (<20 %) (JP 6833585 B2). An older European example uses 70 % colloidal silica, 5 % ureido-silane and 25 % nonionic surfactant, on a solids basis (EP0500073A2).

Functional-particle versions keep the removable base recipe but replace part of the CaCO3. Interference pigments such as Iriodin SHR 9870/Solarflair (TiO2–SiO2–TiO2 on mica) reflect NIR. Scatterers that absorb little light give diffusion. A study used Solarflair 870 in a shading paint. It reduced NIR absorption and let more PAR through (ResearchGate). The haze of ReduHeat versus ReduFuse IR (41 % vs 72 % at 22 buckets/ha) shows that one base recipe can balance NIR reflection against diffusion (HortiDaily).

Table 4. Basic recipes taken from patents

Recipe	Main components (typical amount)	Process	How applied	Source
A. Anti-drip liquid	Colloidal silica 3–8 wt%; binder 1–3 wt% (filler:binder 1.6–2.5); polyether-siloxane or acetylenic-diol surfactant 1.5–3.5 phr; optional silane; water ± IPA/EtOH; total solids 5–10 %	Gentle mixing at controlled pH	Factory: 0.1–2 µm dry at 70 °C × 1 min. Field: fine mist on a clean, dry inner surface	JP 6833585 B2; EP0500073A2
B. Removable shading concentrate	CaCO3 (± TiO2) ~60–65 wt%; alkali-soluble/carboxylated acrylic dispersion ~15–20 %; ammonia ~0.4 %; HEC/xanthan thickener 0.1–5 %; SHMP dispersant 0.1–0.5 %; amino-silane 0.05–1 % (RSB only); biocide/defoamer	High-speed mixing, let-down (adding the rest), filter, fill	Diluted 1:2–1:12 on site; 2–9 µm dry	EP 0999736 B1; EP 2045302 B1
C. NIR / diffuse version	Recipe B with part of the CaCO3 replaced by interference pigment (NIR) or scatterers (diffuse)	Flat pigments added with gentle mixing in the let-down	As B	US 7410685 B2
D. Remover, older type	NaOH 2–10 %, NTA/EDTA 2–10 %, benzyl alcohol 10–30 %, xanthan, sulfonate; diluted 5–10×	Mixing	Spray, then rain washes it off within days	EP 0999736 B1
E. Remover, weak-base type	Na2CO3 0.3–7 % at pH 9–10; PVA/HEC thickener; no chelant or solvent	Mixing	30 min–24 h contact, then rinse or rain	EP 2045302 B1
F. NIR-absorber concentrate (research)	Cs0.33WO3 20 / solvent 75 / polyacrylate dispersant 5, milled to 80 nm; ~2 phr in resin	Bead milling	Bar, gravure, spray or dip; no greenhouse product found	EP 1676890 B1

The removers on sale confirm the alkaline method:

- ReduClean is pH 14 and ADR class 8 (dangerous goods for transport).
- ReduClean Eco is pH 10, Ecocert-listed and not ADR-regulated.
- Topclear is pH 13.

All of them make the binder dissolve in water, so rain carries it away (ReduClean TDS; ReduClean Eco TDS; Topclear TDS). GRAB warns that products with oxalic or hydrofluoric acid must not be used on plastic covers. Organic growers use 5 % citric acid (GRAB 2020).

4.2 Raw materials are cheap; the small special ingredients carry the cost and supply risk

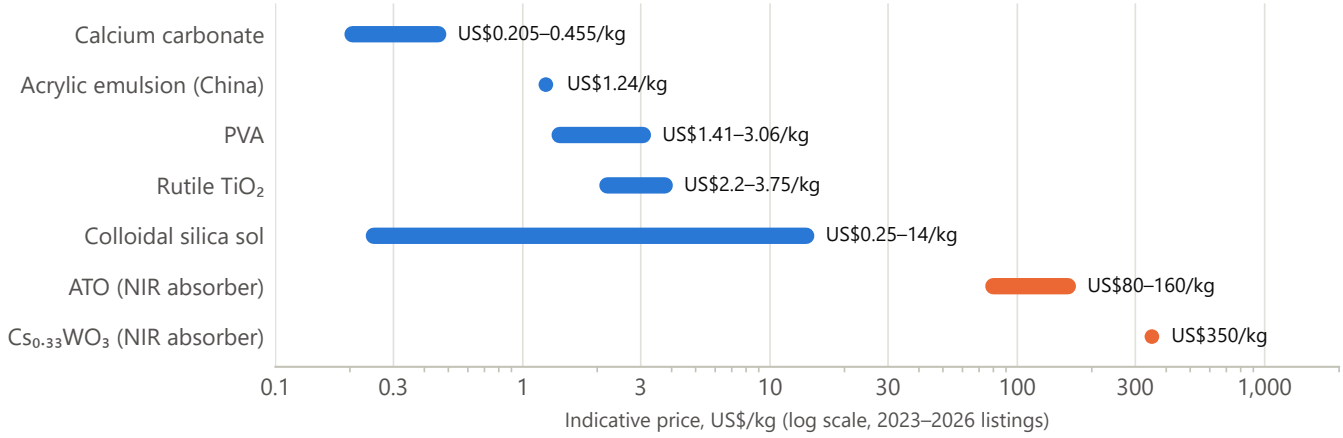


Figure 13. Indicative raw-material prices. Bulk minerals and binders cost cents to a few dollars per kilogram; NIR-absorbing nano-oxides cost two to three orders of magnitude more.

Source: Table 5 (Intratec, SMM, ECHEMI, IndiaMART, ChemicalBook, China Tungsten, Accio listings, 2023–2026).

Bulk ingredients cost US$0.2–4/kg. Special particles are sold only on quote, or at catalogue prices 100 to 100,000 times higher. This big difference explains two things. The cheap end of the market (Chinese chalk paints) is very cheap. And premium NIR products cost about four times as much as white shading.

Table 5. Raw materials: function, amount and rough prices

Component	Function	Typical amount	Example suppliers / grades	Rough price (date, basis)
Calcium carbonate	Shading pigment	30–95 wt% of concentrate	Omya (named in a patent)	US$205/t Europe, 239/t NE Asia, 455/t US (Nov 2025, generic)
Rutile TiO2	White pigment with high refractive index	Part of pigment share; TiO2 grades diluted 5–12×	Longbai (price setter)	China RMB 15,500–17,000/t (Apr 2026); ~US$2,210/t ex-works export (May 2026); Europe US$3,750/t (Nov 2025)
Acrylic emulsion	Removable binder	5–60 wt%	NeoCryl B817/FL711; Wanhua, Baolijia, BATF	China US$1,240–1,261/t (H1 2026); up to +15 % announced Aug 2026
PVA	Water-loving anti-drip binder	0.5–5 wt%	–	US$1,410–3,062/t (Q4 2025–Q1 2026, by region)
Colloidal silica sol	Anti-drip network	2–12 wt%	Nissan Snowtex, Cataloid SI-35, LUDOX	India 30 % bulk ₹21–68/kg; China small packs RMB 85–100/kg (Apr 2026); catalogue $81/L
Alumina sol / boehmite	Positively charged co-colloid	10–15 wt% sols	Sasol Disperal P2	Not found
Interference pigment	NIR reflection	Not disclosed	Merck Iriodin SHR 9870 / Solarflair	Not found
Cs0.33WO3	NIR absorber	~2 phr; 0.9 mg/cm²	Sumitomo Metal Mining (original maker)	US$350/kg coarse powder (Feb 2023); nano retail €509/25 g
ATO	NIR absorber / antistatic	20–40 % dispersions	Nanoshel, Nano-D	US$80–160/kg B2B listings
UV absorbers / HALS	UV control	0.1–10 phr	BASF Tinuvin, ADEKA LA-81, Clariant Hostavin	Not found
Surfactants, silanes, thickeners, biocides	Wetting, sticking, flow, preservation	0.05–15 phr	Polyether siloxanes, acetylenic diols, HEC, SHMP, CMIT/MIT	Not found

Sources: Intratec CaCO3; SMM TiO2; Intratec TiO2; Expert Market Research acrylic emulsion; ECHEMI; Intratec PVA; Procurement Resource PVA; IndiaMART; ChemicalBook; Sigma-Aldrich; Knowde/Sasol; China Tungsten; Nanografi; Accio ATO; JP 6833585 B2.

These prices show clearly where the margin is:

- The chemistry researcher estimates the raw-material cost of a 63 % CaCO3 / 16 % acrylic shading concentrate at about **US$0.4–0.6/kg before additives and packaging**.
- ReduSol sells at **US$6.4–12.5/kg** in 20 kg pails (Cannavista; Hydrobuilder).
- Sun Clear sells at about US$45/L (Greenhouse Megastore). Its raw materials probably cost well under US$2/L. This last figure is a guess that we could not check, because we found no prices for the special sols and surfactants.

4.3 Production is normal water-based paint practice

The process needs no chemical reactor, unless the plant makes its own sols. For pigmented coatings, CaCO3/TiO2 is first mixed in a high-speed disperser:

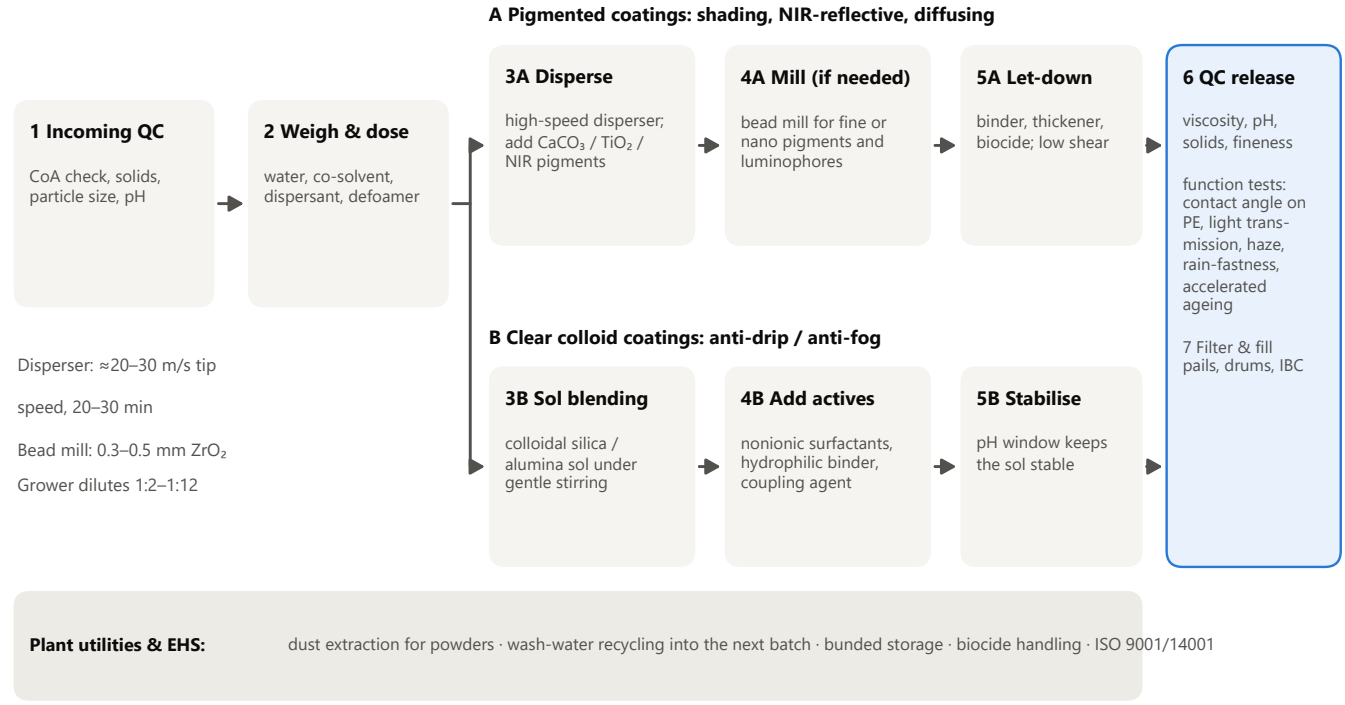


Figure 14. Production flow for water-based functional coatings. Pigmented shading/NIR grades and clear colloid anti-drip grades share intake, QC and filling steps.

Report authors, based on EP 2045302, JP 6833585 and paint-industry dispersion guidance (PCI, NETZSCH).



Figure 15. Beckman DU 640 UV-Vis spectrophotometer on a laboratory bench.

Photo: TimVickers · Public domain · Wikimedia Commons



Figure 16. Crops growing under a green agricultural shade-net enclosure.

Photo: Sandhiya Rangasami · CC BY 4.0 · Wikimedia Commons

- tip speed about 5,000 ft/min (≈25 m/s);
- mill base (the thick first mix) of 1,500–5,000 cP;
- 20–30 minutes after the last powder is added;
- blade-to-tank ratio of about 0.33, and about 1 HP per 10 US gal of mill base.

Then the batch is let down (finished) with the binder, ammonia, thickener and biocide, filtered and filled (PCI practical guide).

Nano NIR absorbers need bead milling with 0.3–0.5 mm zirconia beads. In one case a pigment went from d50 0.76 µm to **41 nm in 180 minutes** at 25 % solids (NETZSCH). Milling uses 0.4–8 kWh/kg of energy, depending on the target size (US 10584256).

Making silica sol in-house would add several steps. Sodium silicate goes through ion exchange to silicic acid at pH 1.3–3.0. Then particles grow at 80–110 °C. Then ultrafiltration raises SiO2 from 6–7 % to 30–35 % (US 4343717; US 6747065). The chemistry researcher thinks this does not pay for a 1,000–10,000 t/yr coatings plant. Buying branded sols is better. Flat interference pigments must be added with gentle mixing, because strong mixing breaks the mica and destroys the effect. This is normal practice for effect pigments, but the notes give no source.

Table 6. Process settings by product family

Step	Key settings	Source
HSD (high-speed disperser) pre-mixing (shading mill base)	4,000–6,000 fpm (≈20–30 m/s); 1,500–5,000 cP; blade:tank 0.33 (paint) to 0.5 (paste); blade 0.5–1.5 D above the bottom; 20–30 min after the last addition	PCI; mixers.com

Step	Key settings	Source
Bead milling (NIR nano-absorbers)	ZrO2/YTZ 0.3–0.5 mm; ~8 m/s at 10 °C → D50 100 nm; target <100 nm and <1,500 cP; transparency needs ≤100–200 nm	NETZSCH; Hockmeyer; EP 1676890 B1
Shading let-down and use	Dual-acrylic binder neutralised with ammonia; field dilution 1:2–1:12; rain-safe in ~30 min	EP 2045302 B1
Anti-drip mixing	4 % silica + 2 % binder + 94 % water; 1.5–3.5 phr surfactant	JP 6833585 B2
Factory in-line coating (benchmark)	Blown film → corona/plasma → 0.1–2 µm water-based coat → hot-air dry (70 °C × 1 min) → wind up	CPPIA; JP 6833585 B2
QC anti-drip life	60 °C bath, 15° slope, fail at >30 % drop area; hot fog 40 °C, ×4 ≈ field life	EP 3350264 B1; PMC5371972
QC optics	Haze to JIS K7136 / ASTM D1003; hemispherical PAR/NIR; contact angle ≤35° at start (CN draft)	JP 6833585 B2; ttbz.org.cn

The equipment is cheap, especially from China. Chinese horizontal bead mills (10–100 L) are listed at about **US$3,650–3,700 per set**. Large-flow models cost US$15,000–150,000 FOB (Alibaba; Degold). One Chinese water-based paint line for 1,000–30,000 L batches is listed at **US$22,500**, but it is not clear what it includes (SIEHE).

The QC (quality control) lab is where the money goes. A Chinese haze meter costs about **US$1,928** (emin.asia). An Indian reseller lists a Q-Lab QUV (UV ageing tester) at **₹16.5 lakh** and a Q-SUN xenon chamber at **₹48 lakh** (IndiaMART). Q-Lab itself says a Q-SUN can cost three times as much as a QUV (Q-Lab LU-8009). A serious lab also needs a UV-Vis-NIR spectrophotometer with an integrating sphere, a contact-angle meter, a fog test rig and greenhouses for field trials. We could not find public prices for most of these.

4.4 The main patents have expired; active patents cover binder-cleaner systems and smart films

Table 7. Patents (IP) for spray and coated functions

Patent	Subject	Owner	Key date	Status (estimated from the 20-year rule)
EP 0999736 B1	Removable pigment/acid-binder coating + NaOH/NTA remover	Raadgevend Chemiebureau RSB	Filed 1998	Expired
US 7716872 B2	Coating removable by base + complex former	RSB → Mardenkro	Published 2004	Probably expired or expiring
EP 2045302 B1	Dual-acrylic shade coating + carbonate cleaner kit	Fenzi Belgium	Filed Sep 2008	May be in force until Sep 2028
US 11634590	Removable, biodegradable coating	Not captured	~2023	Probably active
EP0500073A2 / US 5262233	Colloidal silica + silane anti-drip agri film	Not confirmed	Early 1990s	Expired
JP 6833585 B2	Rough silica drip layer with dry/wet haze switch	Not in machine-readable text	Published 2021	Probably until ~2036–37
EP 3350264 B1	In-film PEO–PE anti-drip blend	Dow	Filed 2015	Probably until ~2035
EP 1676890 B1	Cs0.33WO3 NIR-shielding particles	Sumitomo Metal Mining	Filed 2004	Expired Oct 2024
US 7410685 B2	IR-reflective interference pigments for greenhouses	Merck (not verified)	~2008 grant	Probably expired 2024–26
CN112961519A	Long-lasting anti-fog coating liquid for greenhouse film	Not captured	2021	Unknown

A new company can use the general chemistry of removable whitewash, colloidal-silica anti-drip and CsWO3. It should check freedom-to-operate (whether it may use a technology without breaking patents) for specific binder-cleaner combinations, haze-switching rough surfaces and recent Chinese coating-liquid claims. The expiry dates in Table 7 are estimates. They do not include term extensions or lapsed patents.

CHAPTER 5

Application is moving from people on roofs to machines and drones

Coatings succeed or fail in the field. The most common failure is an uncontrolled dose. In Almería, **99 % of growers whiten their covers**. A team of two people uses a hydraulic pump, and one walks on the roof with a hose that has no regulator. They apply about 0.1 L/m² (≈1,000 L/ha) and put on whatever amount "he considers suitable". So the real dose in g/m² is unknown (López-Martínez et al. 2019). The UAL (University of Almería) authors ask for doses to be standardised in g/m² (Moreno-Teruel et al. 2020). A Dutch machine maker says the law often does not allow walking on covers at all (HortiDaily 2021).



Figure 17. Multirotor agricultural drone spraying a rice field during an IRRI demonstration in the Philippines.

Photo: Christopher Hedreyd - PIA 4A CALABARZON · Public domain · Wikimedia Commons



Figure 18. Manual knapsack sprayer with spray lance.

Photo: Safapalanadan94 · CC BY-SA 4.0 · Wikimedia Commons



Figure 19. Self-propelled boom sprayer working in a cereal field near Congham, Norfolk, England.

Photo: Richard Humphrey · CC BY-SA 2.0 · Wikimedia Commons

Table 8. Outside and inside application methods

Method	Speed / team	Spray volume and dose	Evenness and limits	Cost evidence	Source
Hand hose from roof (Almería)	2 people (pump + roof walker)	≈1,000 L/ha; 0.083–0.5 kg/L CaCO3 (8–50 g/m²)	No dose control; depends on wind; risk of falling	€0.09/m² per crop cycle for 2 applications + 1 wash (2005–06)	López-Martínez 2019; Cajamar
Pump set + spray gun (Lumiforte guidance)	n/r	1,700–2,000 L/ha; diaphragm pump ≥40 L/min; +10 % product vs machine	Runs down: thin at ridge, thick at gutter	Depends on product	Grower2Grower; ReduFuse label
Roof-running washer with spray boom (e.g., Besseling Roofmaster)	n/r	1,000–1,500 L/ha; boom ~20 cm above the roof	Most even layer; up to 10 % less product; needed for diffuse/NIR grades	Machine owned or contracted	HortiDaily 2021
Helicopter	"Large areas in a short time"	1,000–1,500 L/ha	Drop pattern; good enough for ReduSol, weaker for diffuse/spectral grades; flying over not allowed everywhere	"Relatively low"	FloralDaily 2022
Drone, XAG trial (Australia)	0.6 ha in ½ day vs 2 days by hand	12 vs 20 barrels of whitewash	40 % less material (article claims 60 %)	≈US$1,700/ha vs ≈US$2,830/ha material (derived)	Campo Digital
Drone service (Germany, DAS)	10 ha coated + 10 ha cleaned in 2023	Narrow, overlapping lines; ReduSol	Safety is the main benefit	n/r	HortiDaily 2024
Drone service (Spain)	"Much less time"	"Less than half the whitewash"	Even; AESA permits affect price	"Similar to traditional"	Drovinci
Drone service (Korea, county-subsidised)	30 of 30 farms that applied chose drones (Ganghwa)	Vinyl houses	No standard methods	≈₩1,100/pyeong (≈₩3.3 M/ha), 60 % county subsidy	전국매일신문; 로컬세계
Farmer-modified drone (Shouguang, China)	160 m house in ~10–15 min vs ≥2 h	8 special nozzles	Single reports only	n/r	CJVeg
Drone contractor (US, Fair Lifts)	Mobile mixing trailer	XAG P100/P150 with RTK; maps of what was applied	FAA Part 107 + 137 + Exemption 44807	Price on quote	Fair Lifts

Method	Speed / team	Spray volume and dose	Evenness and limits	Cost evidence	Source
Inside pipe-rail cart (AntiCondens)	Cart moves 40 m/min	60 L product in ~1,000 L/ha	Empty, dry greenhouse; must dry before condensation	Product + rental + labour	HortiDaily 2019

Drone hardware sets the limits. A DJI Agras T50 carries 40 L and sprays 16–24 L/min. Its drops are 50–500 µm, it covers a 4–11 m strip, and it weighs 92 kg at take-off when spraying (DJI T50). It costs about **€12,800–13,000 including VAT** in Spain (Cultivotech 2026). Ground machines use 1,000–2,000 L/ha. At that volume, a T50 would need 25–50 tank loads per hectare. So drone operators must use strong, low-volume mixes. This is why Lumiforte asks growers to check mixing ratios for each drone (Greenhouse Grower), and why it launched the drone-ready Eclipse LD+ (Eclipse LD+ TDS).

Regulation is the main brake. Dutch rules allow only two take-offs per day per site under general exemptions. A 2026–2027 pilot across provinces allows unlimited daily flights for electric drones of 25–150 kg. Its picture shows a drone whitening greenhouses (Zuid-Holland/IPO). The DJI T100 weighs 175 kg when spraying, which is above that limit (DJI T100).

5.1 Short weather windows and removal by rain shape the product design

Sellers agree on what a coating needs:



Figure 20. Tomato crop with drip irrigation inside a plastic polytunnel at Sézegnin, canton of Geneva, Switzerland.
Photo: MHM55 · CC BY-SA 4.0 · Wikimedia Commons



Figure 21. Strawberries on plastic-mulched beds and leafy greens inside a plastic tunnel in North Macedonia.
Photo: Иван Ж · CC BY-SA 4.0 · Wikimedia Commons

- The cover must be clean and dry. The "oil layer" must be removed from new film.
- RH (relative humidity) below 80 %.
- About 10–25 °C (5–25 °C for ReduSol).
- No rain for at least 12 hours.
- Wind no more than 7 km/h (Grower2Grower/ReduSystems).

The table below shows the application windows for each product.

Table 9. Application windows by product

Product	Temperature (°C)	No rain after application	Drying	Wind
ReduSol	5–30	6 h	5–6 h at 20 °C	<2 m/s
ReduHeat, ReduFuse, ReduFuse IR	10–25	12 h	–	<2 m/s
Transpar, Eclipse LD/LD+	10–30	6 h	–	<2 m/s
Optifuse / Optifuse IR	10–30	12 h	–	<2 m/s
Eclipse F4, Canope F4	5–30	12 h	–	<2 m/s
AntiReflect	10–20	24 h	~2 h	<2 m/s
D-Fuse	>5, dry	No rain expected	–	–

Sources: Lumiforte TDS linked in Table 2; Royal Brinkman D-FUSE.

Removal also depends on the weather. ReduClean is sprayed at least at 250 L/ha onto a dry roof "when rainfall is expected in the next few days". Diffuse products need a lot of rain within two days, and a brushing machine ready to use. Roof sprinklers make removal harder (ReduSol information sheet; ReduFuse label). Different climates need opposite things:

- In Almería, a whitewash without glue was applied on 19 August 2014. It washed off with 12.6 mm of rain on 1 October (Moreno-Teruel et al. 2020).
- In Korea, coatings that rain washes off after 2–3 months need no removal work (RDA via korea.kr).

So the product design differs by climate. Monsoon Asia needs sticky products that wear off by themselves. Dry regions need products that machines can remove, or that need no rinsing.

Seasonal plans are clear. In Almería, the advice is 35 g/m² (0.5 kg/L) at August transplanting. It is washed off when the inside maximum falls below 35 °C. Then 15 g/m² goes on in late spring (Moreno-Teruel et al. 2020). Dutch growers put ReduHeat or ReduFuse on early and add a thin ReduSol layer later. ReduSol must always be the top layer (HortiDaily). Inside anti-condensation goes on when the crop is changed. It damaged roses in seller tests, and cleaning agents remove it (HortiDaily 2019). Chinese solar greenhouses have one more problem: straw mats or nets pulled over the film rub the coating off (191.cn).

CHAPTER 6

Field results: 2–5 °C cooling is real; yield claims above 10 % are not



Photo: Schumi4ever · CC BY-SA 3.0 · Wikimedia Commons

Independent evidence shows a modest but reliable benefit. Larger claims come almost only from sellers, local governments, grant forecasts or very small test greenhouses. A chart could compare peer-reviewed results (2–5 °C cooling, 2–6 % yield) with press and seller claims (3–10 °C cooling, up to +34 %). It would make the gap easy to see.

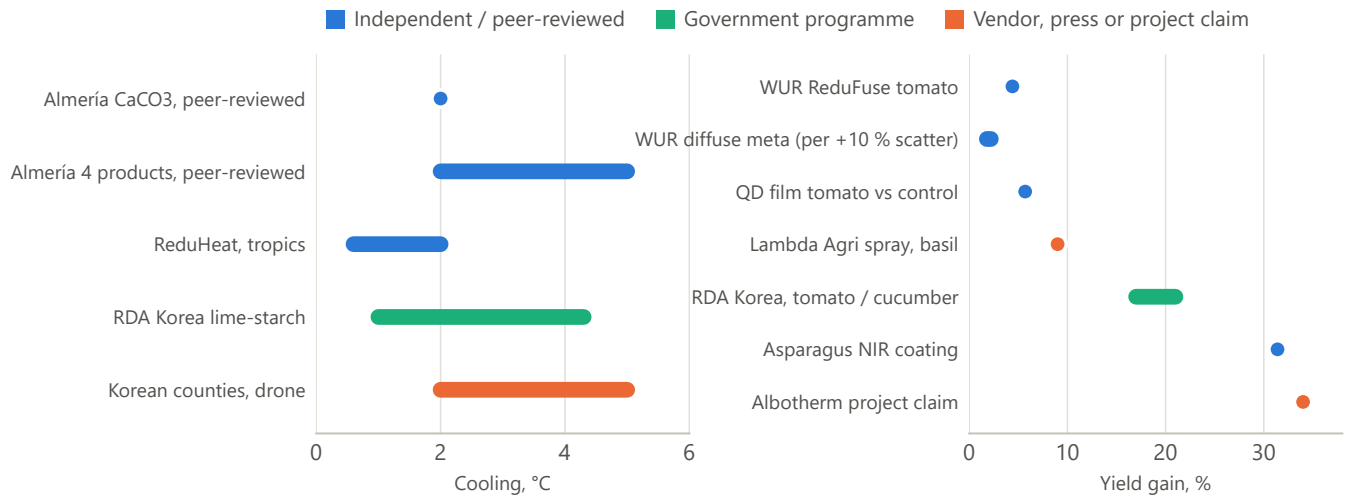


Figure 22. Measured effects versus claims. Independent trials cluster at 2–5 °C of cooling and single-digit yield gains; double-digit gains come mainly from government programmes or vendor and project claims.

Source: Table 10 (Moreno-Teruel 2020, López-Martínez 2019, WUR 2011 and 2025, RDA, Korean county press, Lambda Agri, Albotherm).



Figure 23. Interior of a polyethylene high tunnel with vegetable crops at a farm in Indiana, USA (2022).

Photo: USDAgov · Public domain · Wikimedia Commons



Figure 24. Polyethylene-covered seasonal high tunnel during a field tour in South Dakota, USA (2012).

Photo: USDA NRCS South Dakota · Public domain · Wikimedia Commons

Table 10. Measured field effects of spray-on coatings and similar covers

Study	Place / crop	Treatment	Result	Type of evidence
Moreno-Teruel et al. 2020	Almería, tomato	CaCO3 0.5 vs 0.25 kg/L	August Tmax −2 °C (from 35.0 °C); days >37 °C cut from 9–10 to 3; yield −0.8 to −1 % per −1 % transmission; high dose −4 to −5 % production	Peer-reviewed (Agronomy 10:237)
López-Martínez et al. 2019	Almería, empty house	Four CaCO3 products at 0.25 kg/L	Max ΔT at hottest hour −2.0 to −5.0 °C; more even temperature	Peer-reviewed (IJERPH 16:958)
Gázquez et al. (Cajamar, 2005–06)	Almería, sweet pepper	Whitening vs fog + half whitening	Marketable 10.2 vs 8.4 kg/m²; net income €2.86 vs €1.13/m²; whitening €0.09/m²/cycle	Institutional (Cajamar)
Cajamar 2004 trial	Almería, pepper	Lime 1:6 vs 1:4	Yield 7.6 vs 7.7 kg/m² (not significant); higher dose controlled heat better	Institutional (Cajamar)
WUR 2011 (Dueck et al.)	NL, tomato	ReduFuse 1:8 (≈50 % haze)	Transmission 78 % vs 82 %; PAR sum 93 % of reference; production +4.4 %	Independent (WUR edepot 178476)
WUR meta-analysis 2025	11 crops	Diffuse materials	≈+1.8 % yield per +10 % Hortiscatter (tomato 2.1 %, cucumber 2.2 %)	Research, search snippet (WUR eDepot)
RDA Korea 2017	Korea, vinyl houses	Lime-starch coating	−3–4 °C; tomato +21 %, cucumber +17 % yield; strawberry sugar +4 %	Government (농민신문)
Chen & Shen 2022	Taiwan (presumed), asparagus	NIR-reflective diffusion coating on PE	Yield +31.4 %	Peer-reviewed, via review (Frontiers 2025)
López-Marín et al. 2019	Spain, tomato	Anti-NIR film	Lowest yield of all covers tested	Peer-reviewed, via review (Frontiers 2025)
Tropical study	–	ReduHeat	Max air temperature cut 2 °C in dry season, 0.6 °C in rainy season	Independent, snippet (core.ac.uk)
Van Paassen nursery 2019	NL, tomato, glass	AntiCondens	+7–9 % light vs untreated	Seller measurement (HortiDaily)
Avadan 2021–22	Dagestan, tomato, plastic	AntiCondens Plastic	+15–20 % light	Grower claim (HortiDaily 2022)
Katsoulas et al. 2007	Greece, cucumber	AD + anti-mist film vs AD film	RH ≈80 % vs 95–100 %; half the Botrytis sprays	Linked to manufacturer (IPPS 2009)
Korean counties 2025–26	Vinyl houses	Shading applied by drone	−2 to −5 °C	Press (Incheon Ilbo)
Albotherm trials	UK	Coating activated by heat	+34 % average yield	Project claim (HortiDaily 2026)
Lambda Agri	UK	Sprayed UV→red varnish	Basil +9 %	University press release (Univ. of Bath)

We draw three conclusions:

1. **Whitewash dose is a trade-off with a known "exchange rate".** Each 1 % of transmission lost costs about 0.8–1 % of yield. So high doses pay only during heat peaks, and they must come off quickly (Moreno-Teruel et al. 2020). This is the argument for NIR-selective products, which block heat with less PAR loss.
2. **The NIR case is likely, but not proven on plastic.** There is one negative result for anti-NIR film. And no peer-reviewed study compares NIR coating with CaCO3 on plastic houses. So we must be careful. There are also side effects. ReduHeat removes some far-red light, and this made bedding plants 7–17 % shorter in a French study (HortiDaily).
3. **The value of anti-drip rests on seller data.** The physics is solid. But a consultant claims 80–90 % less fungicide (HortiDaily 2020), and no peer-reviewed study confirms it. A new supplier with independent trial data over several seasons would stand out just by having this data.

CHAPTER 7

A €60–370 m coatings niche sits inside a US$13 bn film market



Photo: Hayan Alhasan · CC BY-SA 4.0 · Wikimedia Commons

7.1 Film markets are large, but nobody measures the coating market

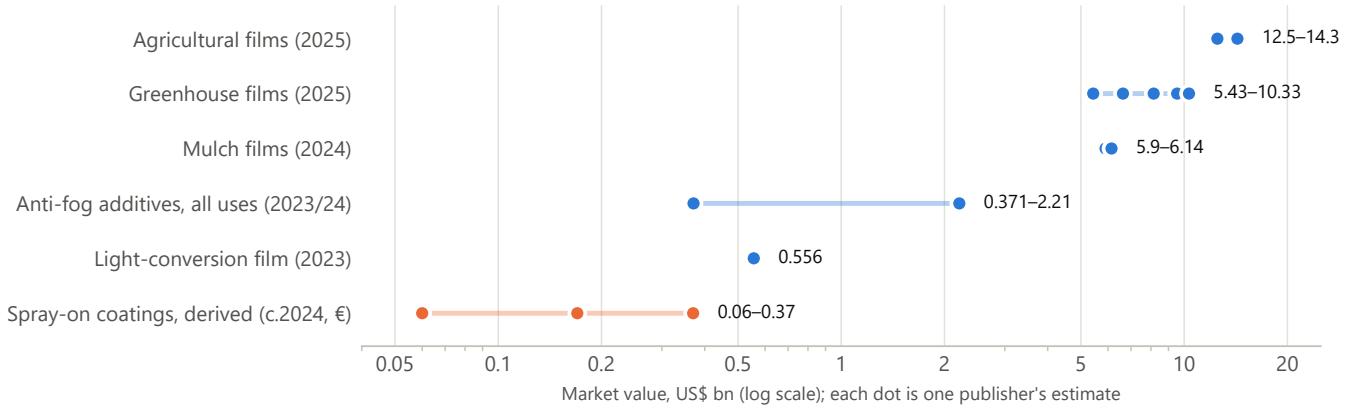


Figure 25. Published market-size estimates and the derived coatings niche. Estimates for the same segment vary up to sixfold; spray-on coatings are about 1 % of the agricultural-film market.

Source: Table 11 (publisher teaser figures; scope undisclosed) and Table 12 (bottom-up estimate by the report authors).

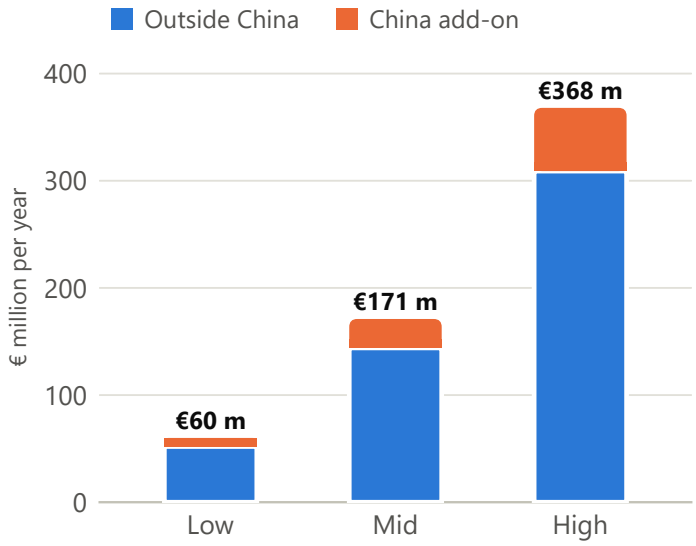


Figure 26. Bottom-up scenarios for spray-on coating materials. Treated shares and spend per hectare are assumptions; see Table 12.

Source: report authors, from Tong et al. 2024 area data and Lumiforte cost points.

Table 11. Published market-size estimates (summary figures; scope and method not disclosed)

Segment	Publisher	Year	Value (US$ bn)	CAGR (yearly growth)
Agricultural films	Fortune BI	2025 / 2034	12.72 / 21.75	6.0 % (2026–34)
Agricultural films	Mordor	2025 / 2031	12.50 / 18.73	6.98 %
Agricultural films	GMI	2025 / 2035	14.30 / 25.13	5.8 %
Agricultural films	Grand View	2023 / 2030	11.45 / 20.59	8.8 %
Greenhouse films	GMI segment / TBRC / Fortune BI / EMR / Reanin	2025	5.43 / 6.63 / 8.15 / 9.54 / 10.33	8–10.6 %
Mulch films	Market.us / Fortune BI	2024	5.9 / 6.14	8.9 %
Mulch films	Grand View	2030	7.68	10.7 %
Anti-fog additives (all uses)	GMI / Grand View	2023 / 2024	0.371 / 2.21	~4.9 %
Agricultural light-conversion film	Bosson	2023 / 2029	0.556 / 0.812	6.5 %
"Greenhouse coating" (scope unclear)	Reports and Data	2024 / 2034	1.2 / 2.5	7.5 %
Spray-on greenhouse coatings, materials (derived)	**Bottom-up estimate**	**c.2024**	**€0.06 / 0.17 / 0.37 bn (low / mid / high)**	**4–8 %/yr working assumption**

Sources: Fortune BI; Mordor; GMI; GVR via MarketResearch.com; TBRC via GII; EMR; Reanin; Market.us; GVR mulch; GMI anti-fog; GVR anti-fog via GII; Bosson; Reports and Data.

Two warnings:

- Greenhouse-film estimates differ by almost 2×, and anti-fog estimates by about 6×. Grand View's 2024 "actual" value for anti-fog additives is exactly the same as the forecast in its own 2016 press release. This suggests the model was copied forward, not counted again (PR Newswire 2016).
- The US$1.2 bn "greenhouse coating" figure lists big glass and polymer firms (Saint-Gobain, BASF, Dow, SABIC) and no shading-paint maker. Do not present it as the shading-paint market (Reports and Data).

So the bottom-up estimate in Table 12 is the most important one. It starts from the permanent greenhouse area mapped by satellite: at least **1.3 Mha (2019)**, with 60.4 % in China (Tong et al., Nature Food 2024). It multiplies this by an assumed treated share and by spending per hectare. The spending figures come from Lumiforte's own cost points: ReduSol ≈ 0.06/m² and ReduHeat ≈ 0.26/m² (HortiDaily 2023).

Table 12. Bottom-up scenario for spray-on coating materials (derived; treated shares are assumptions)

Scenario	Area basis (ha)	Treated share	Spend (€/ha/yr)	Result (€ m)
Outside China, low	515,000	20 %	500	52
Outside China, mid	515,000	35 %	800	144
Outside China, high	515,000	50 %	1,200	309
China added, low / mid / high	785,000	5 / 10 / 15 %	200 / 350 / 500	8 / 27 / 59
World total, low / mid / high				**60 / 171 / 368**

The estimate does not include three things: seasonal tunnels (satellite maps do not show them), drone and contractor services, and lime itself. Even the mid case is only about 3 % of GMI's greenhouse-film segment. The China line is the least certain. East Asian price data (below) suggest that spending per hectare there is much lower than the assumed €200–500.

7.2 Protected growing area is growing in the Global South, and is mostly plastic

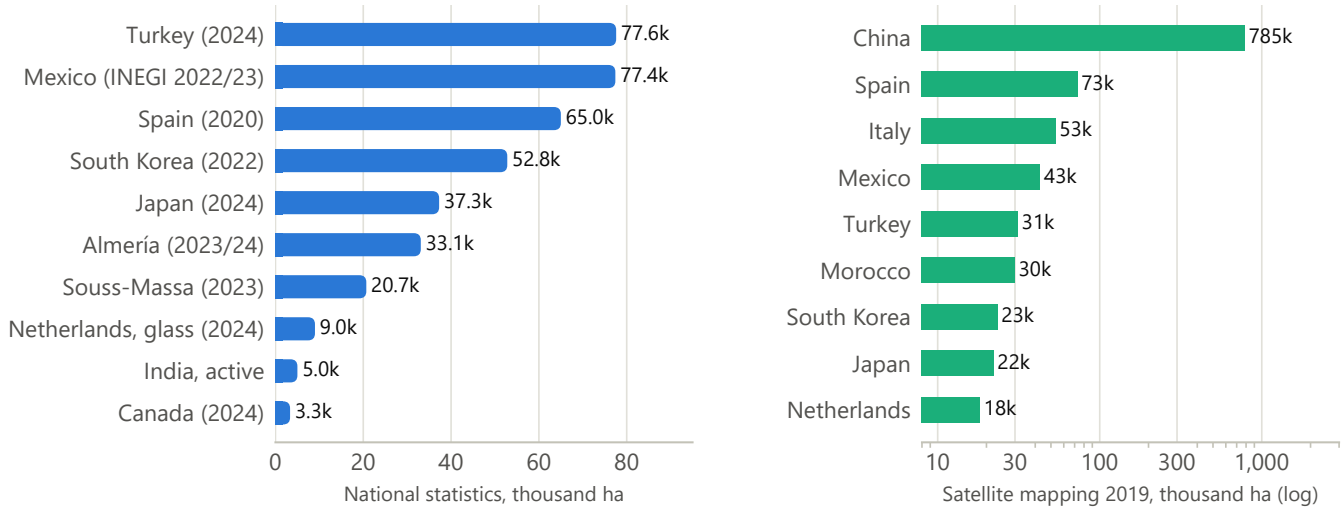


Figure 27. Greenhouse and protected-cultivation area. National statistics (left) include tunnels and differ by definition; satellite mapping (right) counts permanent greenhouses only — do not compare the two directly.

Source: Table 13 (national statistics offices; Tong et al., Nature Food 2024).



Figure 28. Rural road lined with arched plastic-film greenhouses (vinyl houses) in South Korea.

Photo: Abc46224 · CC0 · Wikimedia Commons



Figure 29. Rows of plastic-film vegetable tunnels (dapeng) beside a provincial road in China.

Photo: 琳飞 · CC BY-SA 3.0 · Wikimedia Commons



Figure 30. Extensive field of plastic-film greenhouses at Xichang, Sichuan, China (2023).

Photo: MNXANL · CC BY-SA 4.0 · Wikimedia Commons



Figure 31. Long plastic-film greenhouses (vinyl houses) flanking a road in rural Japan.

Photo: Mamusi Taka · CC BY-SA 3.0 · Wikimedia Commons



Figure 32. Polyhouses on terraced hillsides at village Dhaar, Solan district, Himachal Pradesh, India.

Photo: Harvinder Chandigarh · CC BY-SA 4.0 · Wikimedia Commons



Figure 33. Glass greenhouses with open roof vents at 's-Gravenzande, Westland, Netherlands (2019).

Photo: Steven Lek · CC BY-SA 4.0 · Wikimedia Commons

Table 13. Greenhouse and protected-growing area (do not mix the two data series in one chart)

Country	Satellite series, 2019 (ha)	National statistics (ha, year)	Trend
World	≥1,300,000 (119 countries)	GMI cites 3.37 Mha of plastic greenhouses incl. tunnels	No growth in the North; growth in Asia, Africa, Latin America
China	~785,000 (60.4 %)	~2.85 Mha (42.7 m mu; 69 % plastic tunnels, 29 % solar greenhouses, <3 % multi-span)	Plan: upgrade 2 m mu by 2030
Spain	~72,800	~65,000 (2020 census); Almería 33,100 productive (2023/24)	+0.6 %/yr
Italy	~53,300	–	–
Mexico	~42,900	23,483 (SIAP 2022) to 77,417 (INEGI 2022/23)	–
Turkey	~31,200	77,600 (2024)	+45 % since 2002
Morocco	~29,900	Souss-Massa 20,695 (2023)	+7.7 %/yr since 2015

Country	Satellite series, 2019 (ha)	National statistics (ha, year)	Trend
South Korea	~23,400	52,800 (2022); 99.2 % vinyl	–
Japan	~22,100	37,278 (2024); 53.5 % PO, 32.6 % PVC, 4.2 % glass	−1.7 %/yr
Netherlands	~18,200	>9,000 glass (2024)	~flat
India	–	14,366 built 2005–20, ~5,000 in use	Driven by subsidies
Canada	–	3,330 (33.3 M m², 2024)	+4 % in 2024

Sources: KU/Tong et al.; FreshPlaza; MARA/CAAS 2024; Maldita; Infoagro; Acta Universitaria; East-Fruit; EGU25; 농민신문; MAFF 2026; Glastuinbouw Nederland; NL survey India; Statistics Canada.

In Japan, 95.8 % of houses have plastic covers; in Korea, 99.2 %. China, the Mediterranean and Mexico also have mostly plastic structures. So plastic-covered area is very likely more than 85–90 % of the world's protected area, although no source gives a world split. This plastic area is the target surface for film coatings. Area growth is strongest in hot, dry regions: half of all greenhouse area is in regions with limited resources (KU). These are exactly the climates that need cooling coatings.

7.3 Prices in different regions differ by up to 100 times

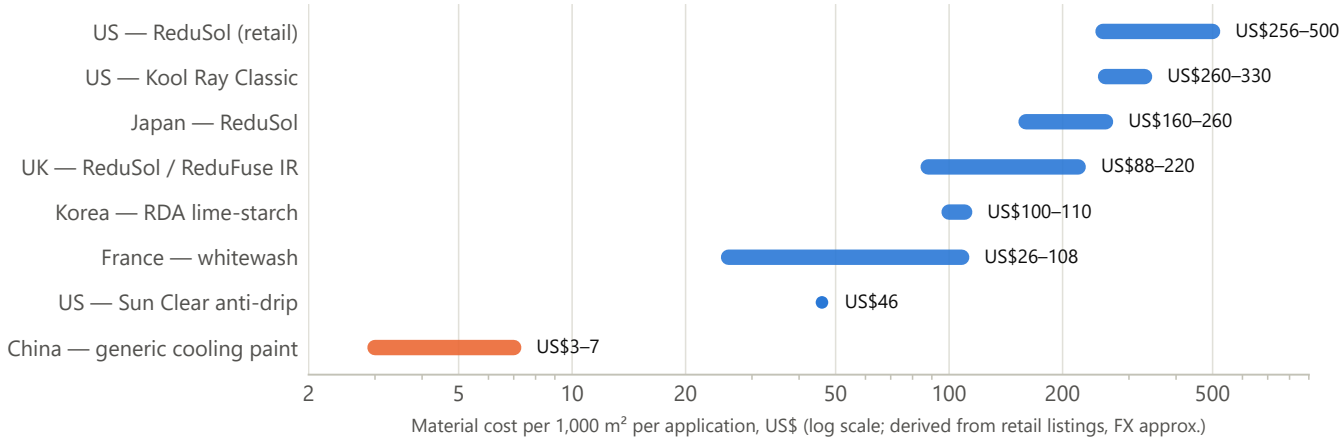


Figure 34. Material cost per 1,000 m² by region and product. Chinese chalk paints cost two orders of magnitude less than imported European brands.
Source: Table 14 (retail listings seen Nov 2026; FX conversions approximate).

Table 14. Shading/heat-coating prices and cost per area by region (shop listings seen Nov 2026 unless dated)

Region	Product	Pack / price	Approx. material cost per 1,000 m² per application	Source
China	立凉 / generic 大棚降温剂 cooling agent	5 kg for CNY 13–35 (≈800–1,000 m²)	CNY 20–50 (≈US$3–7)	惠农网; 链禾网; 1688
Korea	RDA lime-starch coating	KRW ≈150,000 per 1,000 m² (2016)	≈US$100–110	RDA/korea.kr
Korea	Noroo greenhouse shading agent	18 kg for KRW 130,000	–	농달
Japan	ReduSol 20 kg	JPY 24,140–39,600 (≈1 can per 10a)	JPY 24,000–40,000 (≈US$160–260)	吉田化成; おてんとさん
Japan	Hermadix Q4 White / Q-Heat / Sudlac Transpar	JPY 24,178 / 46,178 / 48,378 per can	NIR grades ≈2× white	Monotaro
Japan	Remover (Removit 20 L)	JPY 21,500 (Jan 2024)	+JPY ~21,000 per 10a	福商資販
Europe (UK)	ReduSol 20 kg / ReduFuse IR 15 kg	£69.83 / £139.66 ex VAT	£70–175 (ReduSol at 10–25 buckets/ha)	Dejex
France	Whitewash / photoselective	~€2/kg / ~€8/kg	€24–100 (whitewash)	GRAB 2020
US	ReduSol 20 kg	US$128–250	US$256–500 at 400 kg/ha (top shop price)	Cannavista; Hydrobuilder
US	Kool Ray Classic	US$38–48/gal (5-gal); 1,575 ft²/gal at 1:8	≈US$260–330	Griffin; BG Hydro

Region	Product	Pack / price	Approx. material cost per 1,000 m² per application	Source
Australia	Hermadix Q4 White 20 kg	A$237	–	Garden Superstore
US (anti-drip)	Sun Clear	US$171/gal; ≈3,700 m² per gal	≈US$46	Greenhouse Megastore

Note: per-1,000 m² values are calculated by us; currency conversions use approximate 2025–26 rates.

Each region is different:

- **Europe and North America** are the premium market. Products are removable and often NIR or diffuse. Lumiforte, Hermadix and Continental serve it. The Netherlands and North-West Europe apply by machine on glass. Plastic houses in Southern Europe and Morocco still use mostly lime (Cajamar).
- **China** is the largest by area, but it is a cheap commodity market. Cooling paint costs CNY 15–35 per mu, and is almost certainly a simple mineral mix. Anti-drip renewal is improvised. The Jingzhou agriculture bureau recommends organosilicone spreaders, and a traditional soybean-flour spray lasts 15–20 days (荆州市农业农村局; 中国数字科技馆). China's 2023–2030 plan for facility farming aims to upgrade 2 million mu of old facilities and reach 60 % mechanisation (MARA).
- **Korea** is the most organised market. RDA developed the recipe, and county governments pay 50–60 % of drone application. But anti-drip additives in the film (73 % is PE) wear out in 3–4 months, and there is no KS rule (한국농정신문).
- **Japan** is the premium import market at JPY 15,000–48,000 per 10a, and removers are sold separately. It recycles 68.9 % of its 79,962 t of waste farm plastics. This favours products that rain can remove (MAFF).
- **India** hardly uses coatings. Subsidised film must already be anti-drip and anti-fog by specification. Replacing the cover gets a 70 % subsidy, and cooling uses aluminet nets (Haryana norms).

7.4 A few coating suppliers sit next to many film makers

Spray-on greenhouse coatings are a concentrated niche, probably with fewer than ten important brands in the world. **Lumiforte** was formed when Sudlac (France, founded 1959) and Mardenkro (NL, founded 1986) combined. Mardenkro launched ReduSol in 1991 and sells it in more than 65 countries (AIPH; HortiDaily). The PE firm **Kartesia** invested in September 2024. The amount was not disclosed, and the aim is to fund M&A (buying other companies) and new business areas (Kartesia). Lumiforte now has four plants (NL, FR, DE, CN). The Chinese site finishes concentrate shipped from the Netherlands. About 30 % of its revenue comes from line marking for sports fields (HortiDaily 2026). This suggests the greenhouse market alone is too small for PE-level growth, which fits our bottom-up size estimate.

Hermadix sells only through Royal Brinkman. Its Temperzon T-74 from 1974 is called the first liquid shading agent (Royal Brinkman). **Continental Products** (Kool Ray) serves North America. **Noroo** and regional chalk makers serve Asia. **Agricultural films** are fragmented: the top five firms hold only about 12 %. The leaders are Armando Álvarez (3.5 %), Amcor-Berry (3.0 %) and Plastika Kritis (2.5 %) (GMI). Big film firms and additive suppliers such as BASF, Croda, Clariant, Evonik, Ampacet and Tosaf build anti-fog, UV and light functions into the film. They do not offer sprays.

7.5 Regulation is now the main force changing coating chemistry

Four groups of rules now shape product design.



Figure 35. Discarded plastic bale-wrap film piled at a roadside next to wrapped bales.
Photo: Alan Liefting · CC BY-SA 3.0 · Wikimedia Commons



Figure 36. Newly planted beds covered with black plastic mulch at Umm Haratayn al-Gharbiyah, Syria.
Photo: Hayan Alhasan · CC BY-SA 4.0 · Wikimedia Commons



Figure 37. Field of raised beds covered with white plastic mulch prepared for strawberries.
Photo: Downtowngal · CC BY-SA 3.0 · Wikimedia Commons

Rinse water. The Delfland water board banned the discharge of coating rinse water from March 2026. Dutch growers must collect chalk wash water and let it settle (HortiDaily 2026; Rodi Westland). This pushes R&D towards bio-based coatings whose residue "becomes a fertilizer".

EU microplastics. Regulation 2023/2055 requires instructions for use and disposal from 17 October 2025 (European Coatings). The exception for paints depends on binder particles forming a film that cannot be undone (CEPE guidance). This does not fit well with a coating designed to be stripped and washed off by rain. The notes advise getting legal confirmation. Polymer microsphere diffusers are the most exposed ingredient. Mineral scatterers avoid the problem.

PFAS ("forever chemicals"). The broad EU restriction would cover almost all fluorosurfactants. The rules are expected around 2027–28, with about five years for paints and coatings to comply (PCI 2026). Patents already show PFAS-free wetting with polyether siloxanes and acetylenic diols (JP 6833585 B2).

Recycling. EN 18109:2025 and the new design-for-recycling rules in EN 13206:2025 favour thin, removable coatings without halogens (Genorma). France's ADIVALOR collected a record 102,000 t of farm plastics in 2024. Its target for 2030 is "100 % collected, 100 % recycled" (Réussir).

On the demand side, climate is the strongest driver. Without action, UK greenhouse yields may fall more than 17 % by 2080, and cooling energy may rise at least 25 % (HortiDaily 2026). Supply shocks show why cheap upgrades of existing covers matter. In 2026, Moroccan plastic pellet costs rose by 50 % or more after a storm and problems with Gulf shipping (HortiDaily).

CHAPTER 8

Plants are cheap to build, margins are high, and market access is the problem

8.1 Plant capex is small and margins are like specialty chemicals

Water-based coating plants only blend and fill. Some benchmarks:

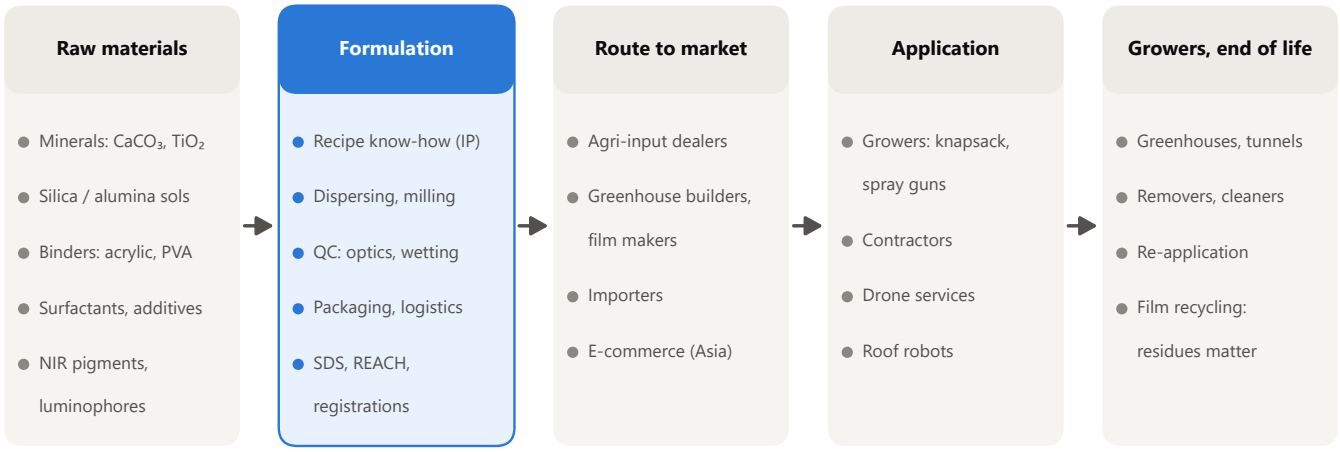


Figure 38. Value chain of sprayable film coatings.
Report authors.

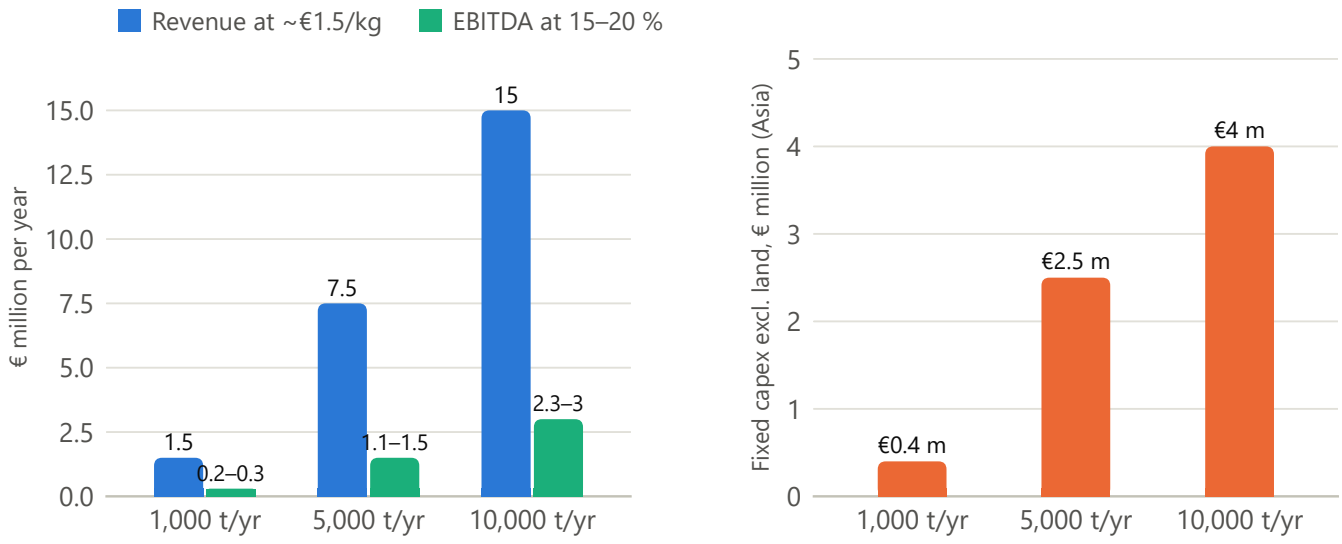


Figure 39. Illustrative plant economics for a new entrant. Capital is modest; the business case depends on winning market share in a concentrated niche.
Source: Table 15 (report authors' illustration from cited benchmarks; not a feasibility study).

- A Hunan EIA (environmental impact assessment) shows a **1,000 t/yr water-based coating project at RMB 2.91 m** total investment. Of this, RMB 0.10 m is for environmental items, on about 1,200 m^2 of land (Yueyang EIA). That is about €360/t of yearly capacity.
- Asian Paints' very large Indore plant costs about €217 m for 400 million L/yr, or about €420/t (European Coatings).
- A full 30-million-L/yr paint plant is estimated at US$15–23 M in an Asian industrial park and US$30–50 M in the US or EU. We could not check the source page of this estimate (Procurement Resource).

For margins, the benchmarks are US specialty chemicals at **35.1 % gross and 18.0 % EBITDA** (Jan 2026) (Damodaran). The agri-film maker Plastika Kritis had 17.3 % group EBITDA on €385 m revenue in 2024 (Plastika Kritis FY2024).

Table 15. Example plant economics for a new company (derived from cited benchmarks; not a feasibility study)

Item	1,000 t/yr	5,000 t/yr	10,000 t/yr	Basis
Fixed capex without land	≈€0.4 m	≈€2–3 m (Asia) / €4–6 m (US/EU)	≈€4 m (Asia)	€360–420/t/yr examples; straight-line scaling of the 30 M L plant benchmark
QC lab + field-trial capability	US$30–60 k (low-cost instruments) to several hundred thousand US$ (Western brands)	Same	Same	Haze meter, QUV, Q-SUN prices cited above
Raw-material cost, shading concentrate	≈US$0.4–0.6/kg before additives and packaging	Same	Same	CaCO3 + acrylic prices (Table 5)
Grower price implied by seller example	≈€1.5/kg (600/ha at 400 kg/ha)	Same	Same	HortiDaily 2023; Fair Lifts
Revenue at ≈€1.5/kg	≈€1.5 m	≈€7.5 m	≈€15 m	Before distributor margin
EBITDA at 15–20 %	€0.2–0.3 m	€1.1–1.5 m	€2.3–3.0 m	Specialty-chemical benchmark
Simple payback on fixed capex	≈1.5–2 yr	≈2–4 yr	≈1.5–2 yr	Without working capital and market-entry cost
Share of the €170 m mid-case market	≈1 %	≈4–5 %	≈9 %	Mid-case value at ≈€1.5–2/kg ≈ 85–110 kt

Table 15 is the author's example, built on the benchmarks in the research notes. It makes the main investment point clear. **Money is not the barrier; access to the market is.** A 5,000 t/yr plant pays back in two to four years at benchmark margins. But this works only if it wins about 5 % of a world niche led by one PE-backed company that growers have trusted for 35 years.

Other points:

- **Working capital is seasonal.** In the Northern Hemisphere, application runs March–June and removal September–October. The researchers assume 25–40 % of sales sit in stock and unpaid invoices at peak. This assumption has no source, so it is a planning risk.
- **Freight is a fixed cost problem.** Water-based product in 20 kg buckets has low value per kilogram. This is why Lumiforte ships concentrate to China and finishes it there (HortiDaily 2026).
- **Business models** in the market include distributor sales and exclusive partnerships (Hermadix/Royal Brinkman), concentrate plus local contract finishing, drone-as-a-service (Fair Lifts with Dalsem; AgroDrone Europe) and technology partnerships (Albotherm with Lumiforte).

8.2 Growers earn a strong return on high-value crops, but only a small one on low-value crops

The grower ROI (return on investment) formula is: ROI = (Δyield % × yield/m² × price − Δcost/m²) ÷ Δcost/m². Lumiforte's own example is a Dutch-type tomato crop at 50 kg/m². Changing from ReduSol to ReduHeat adds US$0.20/m². A "very conservative" 2 % yield gain is worth US$1/m². This gives a stated **400 % ROI** (HortiDaily 2023, sponsored). The same maths for Almería pepper (7.7 kg/m², assumed €1/kg) gives only about €0.15/m² for a 2 % gain. That is still more than a basic €0.06/m² whitewash. But it is only just enough against a premium NIR coating at €0.26/m². This explains why lime still leads in low-margin Mediterranean farming.

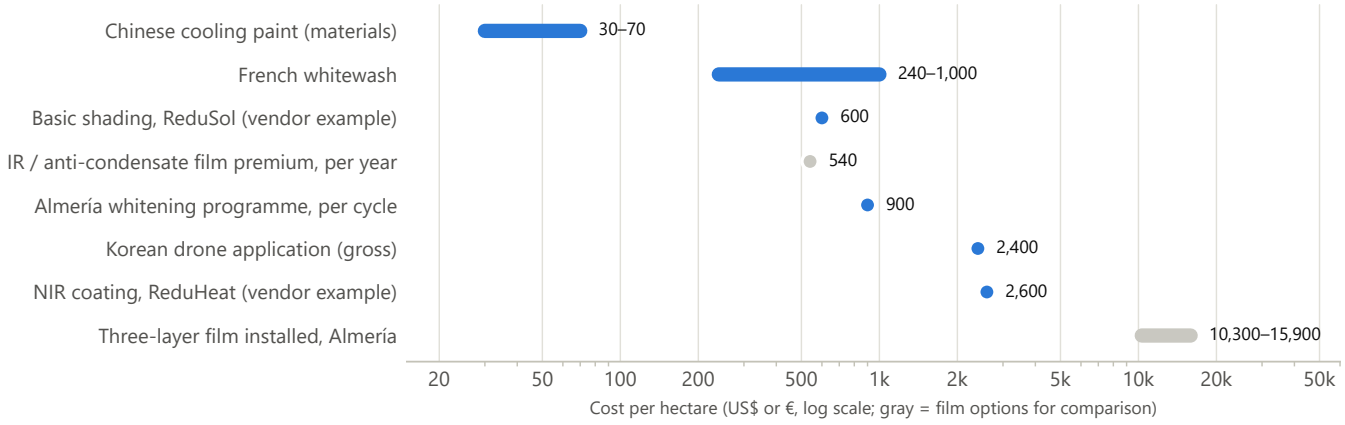


Figure 40. Cost ladder per hectare: coatings versus film options. Coatings sell on seasonality and reversibility rather than lowest cost per m².
Source: Table 16.

The strongest independent economic result is the Cajamar trial. Whitening cost **€0.09/m² per cycle** and gave net income of €2.86/m², against €1.13/m² for fogging. This made it the most profitable cooling option (Cajamar).

Table 16. Cost ladder per hectare: coatings compared with film options

Item	Value	Basis	Source
Chinese cooling paint, materials	≈US$30–70/ha per application	CNY 20–50 per 1,000 m²	惠农网
French whitewash	€240–1,000/ha	~€2/kg	GRAB 2020
Basic shading (ReduSol), seller example	≈600/ha per application (currency not stated)	0.06/m²	HortiDaily 2023
Almería whitening programme	€900/ha per cycle (2005–06)	2 applications + 1 wash	Cajamar
Korean drone application (county programme)	≈₩3.3 M/ha (≈US$2,400), 60 % subsidised	₩1,100/pyeong	전국매일신문
NIR coating (ReduHeat), seller example	≈2,600/ha per application	0.26/m²	HortiDaily 2023
Extra cost for IR/anti-condensate film	≈US$2,150/ha once (≈US$540/ha/yr over 4 years)	"a couple of cents per ft²"	UMass
Three-layer film installed, Almería (2015)	≈€10,300–15,900/ha	€1.03–1.59/m²	Junta de Andalucía
Chinese PO anti-drip film / light-conversion film	CNY 1.05–2.5 / 2.8–3.6 per m²	Shop price	echinagov

The cost ladder shows why coatings sell on being **seasonal, added to existing covers, and reversible**, not on price per m². Basic shading costs about 10–30 % of the yearly film cost, and growers can remove it in autumn. A functional film is cheaper per m², but its function stays all year.

8.3 Venture capital goes to light conversion and smart covers, mostly in small amounts

Table 17. Funding and deals in greenhouse cover and coating technology

Company / project	Date	Amount	Type	Focus	Source
UbiQD	2020 / Apr 2025 / Nov 2025	US$7 m / US$20 m / US$6 m	Series A / Series B / term loan	Quantum-dot light-conversion film	AgFunderNews; HortiDaily
Lumiforte	Sep 2024	Not disclosed	PE (Kartesia)	Coatings platform, M&A	Kartesia
Albotherm	2023 / Feb 2026	£1.6 m / £2 m project	Equity / Defra-Innovate UK	Shade coating activated by heat	Univ. of Bristol; HortiDaily
ENLIGHT (CASCADE)	2020–23	€2.04 m EU of €3.01 m	EIC Accelerator	Light-shifting masterbatch	CORDIS
Voltiris	n/a	CHF 1.4 m + 4.8 m	Pre-seed + seed (incl. 3M Ventures)	Greenhouse solar panels that split the light spectrum	Venturelab
Infrascreen	n/a	US$1.5 m	Seed	Nanomaterials that keep long-wave IR in	Startupticker
i2Cool	May 2024	~HK$100 m	Series A	Radiative cooling (buildings)	CityU

Deals are usually US$1–20 m. They often combine a grant with a strategic investor. Strategic and grower investors matter more than general VCs. Examples are Horticoop and a large tomato producer in UbiQD's round, and R&L Holt in the Albotherm project. The likely exits are sales to big coating or film firms or to PE, not stock market listings (IPOs).

8.4 Research focus has moved from anti-fog to light spectrum, heat and energy covers

OpenAlex (a database of research papers) counts show where research is going:

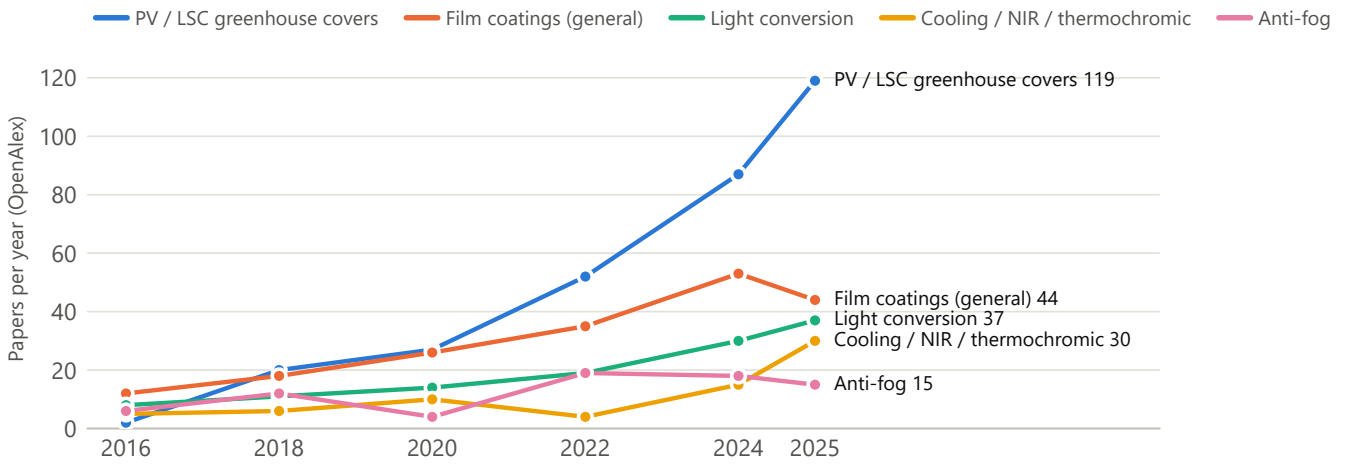


Figure 41. Research publications per year by topic. Energy-generating and spectrum-shifting covers are where research is accelerating; anti-fog is flat.

Source: OpenAlex counts (indicative, ±15 %); Table 18.

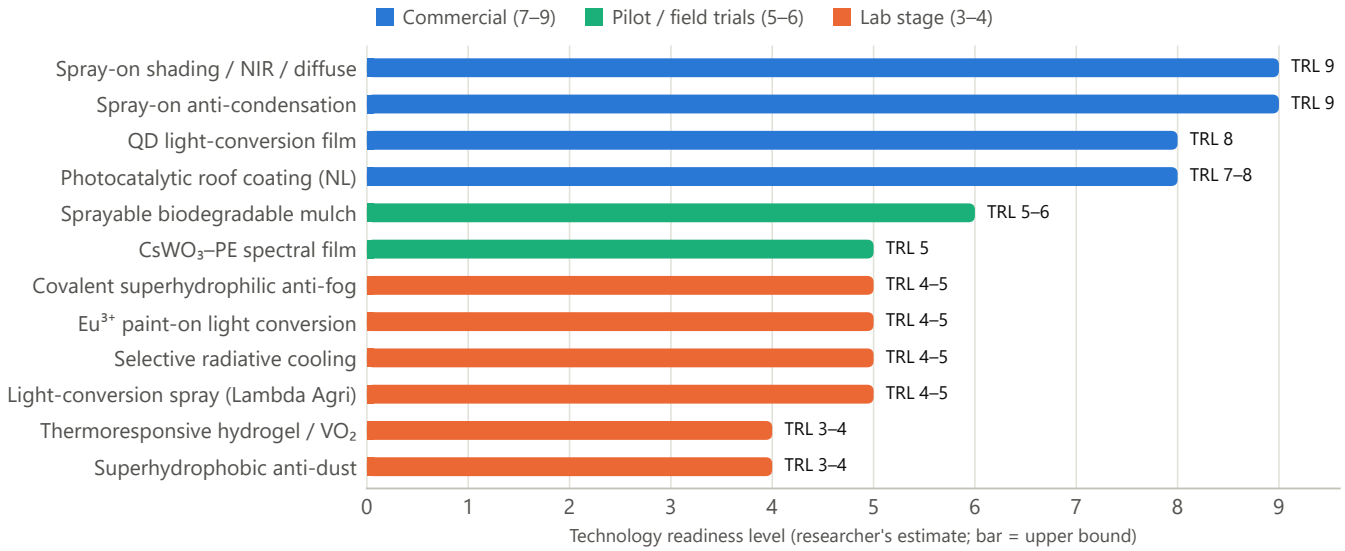


Figure 42. Technology maturity of emerging functions.

Source: Table 19 (TRL estimates by the researchers).

- Papers on coatings for farm, greenhouse and mulch film rose from about 13 a year (2010–19) to 34 a year (2020–25). The peak was 53 in 2024.
- Light-conversion papers grew from 8 (2016) to 37 (2025).
- Papers on PV/agrivoltaic covers (solar power with farming) grew from 20 (2018) to 119 (2025).
- Anti-fog stays a niche with 15–19 papers a year.
- China writes 29–54 % of papers in every topic (OpenAlex API).

The counts are approximate (about ±15 %), and 2026 figures are year-to-date.

Table 18. Papers per year by topic (OpenAlex; 2026 = to 12 Nov)

Year	Film coatings	Anti-fog	Light conversion	Cooling / NIR / thermochromic	PV / LSC greenhouse	Sprayable mulch
2016	12	6	8	5	2	5
2018	18	12	11	6	20	1
2020	26	4	14	10	27	7
2022	35	19	19	4	52	8
2024	53	18	30	15	87	15
2025	44	15	37	30	119	11
2026 YTD	46	15	51	24	164	13

Table 19. How ready new functions are (TRL is the researcher's estimate)

Technology	Best reported result	TRL	Main barrier	Expected timing
Spray-on NIR/whitewash, incl. bio-based	On sale; starch LDC 101 launched 2024	9	Rinse-water rules, removability	Now; residue-free grades 2026–30
Spray-on anti-condensation (surfactant)	"A couple of years" life (extension service)	9	Washes off, must be reapplied	Now
QD light-conversion film	+5.7 % saleable tomato at −14 % DLI (25 weeks, NL)	8	Cost, repeat results	Now; scale-up 2025–28
Self-cleaning roof coating (light-driven)	Launching in NL, no data	7–8 (not verified)	Evidence	Now (NL)
Chemically bonded, very water-loving anti-fog	Contact angle <2°, no drops for 1 month at 60 °C	4–5	Needs corona/plasma; field proof	2026–29, films coated in the factory line
Eu3+ paint-on light conversion	Swiss chard 1.4× biomass in winter, no effect in summer	4–5	Stability under light	2028+
CsxWO3-PE spectral-control film	−25 °C in mini-greenhouses (KSA)	5	Scale-up	2027–30
Selective radiative-cooling laminates	−3 to −5 °C; PAR 37–57 %	4–5	PAR loss	2028–31
Temperature-responsive hydrogel / VO2	Solar T 85 % → 15 %	3–4	Fixed switch temperature, UV ageing	2030+
Water-repelling anti-dust	>145° contact angle, >95 % T (PV glass)	3–4	Wear, PFAS	2030+
Sprayable biodegradable mulch	6-month mulching; same farm results as LDPE	5–6	Results vary in the field	2027–30 (CSIRO target 2027)

Sources: Frontiers in Chemistry 2022; Phys.org/Hokkaido; EurekAlert/KAUST; Mahidol; European Coatings; RSC Mater. Horiz. 2026; Coatings 2021; Sartore et al. 2018; CSIRO; HortiDaily 2026 Multipurus.

Many large yield claims for new covers compare against no cover at all, or come from very small greenhouses in extreme heat. Examples are +152 % for carbon-dot films and "~200 %" for KAUST's film (J. Mater. Chem. C 2025). Against a proper control film, quantum-dot conversion gives about +6 % to +13 %. Green light also helps photosynthesis in the plant canopy a lot. This weakens the idea of converting green light to red (PLOS ONE 2023). Also, a sprayed conversion layer on the outside of a film would get the strongest UV. So photobleaching (fading under light) is the design problem to solve before sales can start.

CHAPTER 9

Conclusion

A spray-on coating can turn normal plastic film into a functional cover. But today this works only where reversibility is the goal: seasonal control of light and heat. Other functions are often listed as film upgrades: UV control, light conversion, long-lasting anti-fog and anti-dust. These belong in the film for physical reasons, not business reasons. They need years of life, protection from UV and wear, and surface activation (corona or plasma treatment) that a field sprayer cannot give.

So the investment opportunity is not a broad "functional-coating" platform. It is a set of more focused bets:

1. A film-safe anti-drip renewal spray that works on PE, EVA, PO and PVC. It would target Asia's millions of hectares of film whose additive wears out in 3–6 months.
2. Shading concentrates made for drones, sold with tested low-volume methods. They would target regions where counties already pay for the flights but have no standard.
3. A mid-priced NIR-selective product that fills the price gap of about 50× between Chinese chalk and European imports.

Regulation is slowly reducing the advantage of the current leaders. Rinse-water bans, microplastics and PFAS rules, and design-for-recycling standards turn the industry's 1990s chemistry (stripping with alkali) into a weakness. They reward mineral or starch binders, removers with only carbonate, and ideas where the residue becomes fertiliser. A new company could design for these rules from the start, but the leader is already moving the same way. The key asset is not plant capacity, which costs only a few hundred euros per tonne. It is independent field data over several seasons, in a market where almost every headline number still comes from the seller.

Appendix: glossary

Term	Meaning in this report
Agricultural film	Thin plastic film (usually 80–200 μm for greenhouse covers, 8–50 μm for mulch) made mainly from LDPE/LLDPE, EVA copolymers or multilayer polyolefin (PO) blends.
Anti-drip (anti-condensation)	A surface treatment that makes condensed water spread into a continuous thin layer that drains away, instead of forming droplets that drip onto the crop.
Anti-fog	Prevents a visible mist of micro-droplets on the film surface; in practice the term overlaps with anti-drip, and some markets use it for a separate "fog-free air" effect.
Contact angle (θ)	Angle a water droplet makes with a surface. High θ (≈90°+) means hydrophobic and drop-wise condensation; low θ (below ~20°) means hydrophilic and film-wise condensation.
Diffuse light / haze	Share of transmitted light that is scattered rather than direct. Diffusing coatings raise haze to spread light deeper into the canopy and reduce leaf scorch.
EVA	Ethylene–vinyl acetate copolymer, used in film layers for better clarity, flexibility and infrared retention.
Functional film	A film that does more than transmit light: e.g., anti-drip, diffusing, NIR-blocking, UV-blocking, light-converting or thermal (IR-retaining).
Haze meter / spectrophotometer	Instruments that measure total and diffuse transmission (haze) and wavelength-resolved transmission/reflection of a film or coating.
In-film additive / masterbatch	Functional chemicals compounded into the polymer before extrusion, as opposed to coatings applied to the surface afterwards.
Light conversion	Absorbing one part of the spectrum (often UV or green) and re-emitting it as light plants use more effectively (often red or blue).
NIR	Near-infrared radiation (≈700–2,500 nm): about half of solar energy, which heats the greenhouse but does not drive photosynthesis.
PAR	Photosynthetically active radiation (400–700 nm), the light plants use for photosynthesis.
PO film	"Polyolefin" multilayer film (a common term in Japan and Korea), often with factory-applied anti-drip coatings.
Shading / whitewash	A white mineral coating (typically calcium carbonate) sprayed on the outside of the cover to reflect sunlight and cool the greenhouse in summer.
Surface energy	A material property that controls wetting. Polyethylene (≈31 mN/m) is far below water's surface tension (72.8 mN/m), so water beads up on untreated film.
Thermicity	A film's ability to block long-wave infrared (≈7–14 μm) heat radiation from the greenhouse at night.
TRL	Technology readiness level, from 1 (basic principle) to 9 (proven in commercial operation).
UV-blocking / UV-open	Films or coatings that remove (or deliberately transmit) ultraviolet light, which affects insects, pollinators, fungal sporulation and crop colour.